Neural Computation -- 2016

109 article(s), all issues

1. Multiple diffusion models to compare saccadic and manual responses

MacInnes W.Joseph

(no DOI)

No abstract available.

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2. State-space analysis of granger-geweke causality measures with application to fmri

SoloVictor

(no DOI)

The recent interest in the dynamics of networks and the advent, across a range of applications, of measuring modalities that operate on different temporal scales have put the spotlight on some sign...

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3. Neural quadratic discriminant analysis

PaganMarino, P SimoncelliEero, C RustNicole

(no DOI)

Linear-nonlinear LN models and their extensions have proven successful in describing transformations from stimuli to spiking responses of neurons in early stages of sensory hierarchies. Neural resp...

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4. Active Inference: A Process Theory

Karl Friston, Thomas H. B. FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, Giovanni Pezzulo

https://doi.org/10.1162/neco_a_00912 | pp. 1-49

This article describes a process theory based on active inference and belief propagation. Starting from the premise that all neuronal processing (and action selection) can be explained by maximizing Bayesian model evidence-or minimizing variational free energy-we ask whether neuronal responses can be described as a gradient descent on variational free energy. Using a standard (Markov decision process) generative model, we derive the neuronal dynamics implicit in this description and reproduce a remarkable range of well-characterized neuronal phenomena. These include repetition suppression, mismatch negativity, violation responses, place-cell activity, phase precession, theta sequences, theta-gamma coupling, evidence accumulation, race-to-bound dynamics, and transfer of dopamine responses. Furthermore, the (approximately Bayes' optimal) behavior prescribed by these dynamics has a degree of face validity, providing a formal explanation for reward seeking, context learning, and epistemic foraging. Technically, the fact that a gradient descent appears to be a valid description of neuronal activity means that variational free energy is a Lyapunov function for neuronal dynamics, which therefore conform to Hamilton's principle of least action.

5. The Population Tracking Model: A Simple, Scalable Statistical Model for Neural Population Data

Cian O'Donnell, J. Tiago Gonçalves, Nick Whiteley, Carlos Portera■Cailliau, Terrence J. Sejnowski
https://doi.org/10.1162/neco_a_00910 | pp. 50-93

Our understanding of neural population coding has been limited by a lack of analysis methods to characterize spiking data from large populations. The biggest challenge comes from the fact that the number of possible network activity patterns scales exponentially with the number of neurons recorded ([Formula: see text]). Here we introduce a new statistical method for characterizing neural population activity that requires semi-independent fitting of only as many parameters as the square of the number of neurons, requiring drastically smaller data sets and minimal computation time. The model works by matching the population rate (the number of neurons synchronously active) and the probability that each individual neuron fires given the population rate. We found that this model can accurately fit synthetic data from up to 1000 neurons. We also found that the model could rapidly decode visual stimuli from neural population data from macaque primary visual cortex about 65 ms after stimulus onset. Finally, we used the model to estimate the entropy of neural population activity in developing mouse somatosensory cortex and, surprisingly, found that it first increases, and then decreases during development. This statistical model opens new options for interrogating neural population data and can bolster the use of modern large-scale in vivo Ca[Formula: see text] and voltage imaging tools.

6. A Combinatorial Model for Dentate Gyrus Sparse Coding

William Severa, Ojas Parekh, Conrad D. James, James B. Aimone
https://doi.org/10.1162/neco_a_00905 | pp. 94-117

The dentate gyrus forms a critical link between the entorhinal cortex and CA3 by providing a sparse version of the signal. Concurrent with this increase in sparsity, a widely accepted theory suggests the dentate gyrus performs pattern separation-similar inputs yield decorrelated outputs. Although an active region of study and theory, few logically rigorous arguments detail the dentate gyrus's (DG) coding. We suggest a theoretically tractable, combinatorial model for this action. The model provides formal methods for a highly redundant, arbitrarily sparse, and decorrelated output signal.To explore the value of this model framework, we assess how suitable it is for two notable aspects of DG coding: how it can handle the highly structured grid cell representation in the input entorhinal cortex region and the presence of adult neurogenesis, which has been proposed to produce a heterogeneous code in the DG. We find tailoring the model to grid cell input yields expansion parameters consistent with the literature. In addition, the heterogeneous coding reflects activity gradation observed experimentally. Finally, we connect this approach with more conventional binary threshold neural circuit models via a formal embedding.

7. Analytical Calculation of Mutual Information between Weakly Coupled Poisson-Spiking Neurons in Models of Dynamically Gated Communication

Jonathan Cannon
https://doi.org/10.1162/neco_a_00915 | pp. 118-145

Mutual information is a commonly used measure of communication between neurons, but little theory exists describing the relationship between mutual information and the parameters of the underlying neuronal interaction. Such a theory could help us understand how specific physiological changes affect the capacity of neurons to synaptically communicate, and, in particular, they could help us characterize the mechanisms by which neuronal dynamics gate the flow of information in the brain. Here we study a pair of linear-nonlinear-Poisson neurons coupled by a weak synapse. We derive an analytical expression describing the mutual information between their spike trains in terms of synapse strength, neuronal activation function, the time course of postsynaptic currents, and the time course of the background input received by the two neurons. This expression allows mutual information calculations that would otherwise be computationally intractable. We use this expression to analytically explore the interaction of excitation, information transmission, and the convexity of the activation function. Then, using this expression to quantify mutual information in simulations, we illustrate the information-gating effects of neural oscillations and oscillatory coherence, which may either increase or decrease the mutual information across the synapse depending on parameters. Finally, we show analytically that our results can quantitatively describe the selection of one information pathway over another when multiple sending neurons project weakly to a single receiving neuron.

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8. On the Mathematical Consequences of Binning Spike Trains

Bruno Cessac, Arnaud Le Ny, Eva Löcherbach

https://doi.org/10.1162/neco_a_00898 | pp. 146-170

We initiate a mathematical analysis of hidden effects induced by binning spike trains of neurons. Assuming that the original spike train has been generated by a discrete Markov process, we show that binning generates a stochastic process that is no longer Markov but is instead a variable-length Markov chain (VLMC) with unbounded memory. We also show that the law of the binned raster is a Gibbs measure in the DLR (Dobrushin-Lanford-Ruelle) sense coined in mathematical statistical mechanics. This allows the derivation of several important consequences on statistical properties of binned spike trains. In particular, we introduce the DLR framework as a natural setting to mathematically formalize anticipation, that is, to tell "how good" our nervous system is at making predictions. In a probabilistic sense, this corresponds to condition a process by its future, and we discuss how binning may affect our conclusions on this ability. We finally comment on the possible consequences of binning in the detection of spurious phase transitions or in the detection of incorrect evidence of criticality.

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9. Orientation Histogram-Based Center-Surround Interaction: An Integration Approach for Contour Detection

Rongchang Zhao, Min Wu, Xiyao Liu, Beiji Zou, Fangfang Li

https://doi.org/10.1162/neco_a_00911 | pp. 171-193

Contour is a critical feature for image description and object recognition in many computer vision tasks. However, detection of object contour remains a challenging problem because of disturbances from texture edges. This letter proposes a scheme to handle texture edges by implementing contour integration. The proposed scheme integrates structural segments into contours while inhibiting texture edges with the help of the orientation histogram-based center-surround interaction model. In the model, local edges within

surroundings exert a modulatory effect on central contour cues based on the co-occurrence statistics of local edges described by the divergence of orientation histograms in the local region. We evaluate the proposed scheme on two well-known challenging boundary detection data sets (RuG and BSDS500). The experiments demonstrate that our scheme achieves a high [Formula: see text]-measure of up to 0.74. Results show that our scheme achieves integrating accurate contour while eliminating most of texture edges, a novel approach to long-range feature analysis.

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10. Fast, Accurate Localization of Epileptic Seizure Onset Zones Based on Detection of High-Frequency Oscillations Using Improved Wavelet Transform and Matching Pursuit Methods

Min Wu, Ting Wan, Xiongbo Wan, Yuxiao Du, Jinhua She

https://doi.org/10.1162/neco_a_00899 | pp. 194-219

This letter describes the improvement of two methods of detecting high-frequency oscillations (HFOs) and their use to localize epileptic seizure onset zones (SOZs). The wavelet transform (WT) method was improved by combining the complex Morlet WT with Shannon entropy to enhance the temporal-frequency resolution during HFO detection. And the matching pursuit (MP) method was improved by combining it with an adaptive genetic algorithm to improve the speed and accuracy of the calculations for HFO detection. The HFOs detected by these two methods were used to localize SOZs in five patients. A comparison shows that the improved WT method provides high specificity and quick localization and that the improved MP method provides high sensitivity.

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11. Online Reinforcement Learning Using a Probability Density Estimation

Alejandro Agostini, Enric Celaya

https://doi.org/10.1162/neco_a_00906 | pp. 220-246

Function approximation in online, incremental, reinforcement learning needs to deal with two fundamental problems: biased sampling and nonstationarity. In this kind of task, biased sampling occurs because samples are obtained from specific trajectories dictated by the dynamics of the environment and are usually concentrated in particular convergence regions, which in the long term tend to dominate the approximation in the less sampled regions. The nonstationarity comes from the recursive nature of the estimations typical of temporal difference methods. This nonstationarity has a local profile, varying not only along the learning process but also along different regions of the state space. We propose to deal with these problems using an estimation of the probability density of samples represented with a gaussian mixture model. To deal with the nonstationarity problem, we use the common approach of introducing a forgetting factor in the updating formula. However, instead of using the same forgetting factor for the whole domain, we make it dependent on the local density of samples, which we use to estimate the nonstationarity of the function at any given input point. To address the biased sampling problem, the forgetting factor applied to each mixture component is modulated according to the new information provided in the updating, rather than forgetting depending only on time, thus avoiding undesired distortions of the approximation in less sampled regions.

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12. LSV-Based Tail Inequalities for Sums of Random Matrices

Chao Zhang, Lei Du, Dacheng Tao

https://doi.org/10.1162/neco_a_00901 | pp. 247-262

The techniques of random matrices have played an important role in many machine learning models. In this letter, we present a new method to study the tail inequalities for sums of random matrices. Different from other work (Ahlswede & Winter, 2002 ; Tropp, 2012 ; Hsu, Kakade, & Zhang, 2012), our tail results are based on the largest singular value (LSV) and independent of the matrix dimension. Since the LSV operation and the expectation are noncommutative, we introduce a diagonalization method to convert the LSV operation into the trace operation of an infinitely dimensional diagonal matrix. In this way, we obtain another version of Laplace-transform bounds and then achieve the LSV-based tail inequalities for sums of random matrices.

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13. Improving the Incoherence of a Learned Dictionary via Rank Shrinkage

Shashanka Ubaru, Abd■Krim Seghouane, Yousef Saad

https://doi.org/10.1162/neco_a_00907 | pp. 263-285

This letter considers the problem of dictionary learning for sparse signal representation whose atoms have low mutual coherence. To learn such dictionaries, at each step, we first update the dictionary using the method of optimal directions (MOD) and then apply a dictionary rank shrinkage step to decrease its mutual coherence. In the rank shrinkage step, we first compute a rank 1 decomposition of the column-normalized least squares estimate of the dictionary obtained from the MOD step. We then shrink the rank of this learned dictionary by transforming the problem of reducing the rank to a nonnegative garrotte estimation problem and solving it using a path-wise coordinate descent approach. We establish theoretical results that show that the rank shrinkage step included will reduce the coherence of the dictionary, which is further validated by experimental results. Numerical experiments illustrating the performance of the proposed algorithm in comparison to various other well-known dictionary learning algorithms are also presented.

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14. Learning the Structural Vocabulary of a Network

Saket Navlakha

https://doi.org/10.1162/neco_a_00924 | pp. 287-312

Networks have become instrumental in deciphering how information is processed and transferred within systems in almost every scientific field today. Nearly all network analyses, however, have relied on humans to devise structural features of networks believed to be most discriminative for an application. We present a framework for comparing and classifying networks without human-crafted features using deep learning. After training, autoencoders contain hidden units that encode a robust structural vocabulary for succinctly describing graphs. We use this feature vocabulary to tackle several network mining problems and find improved predictive performance versus many popular features used today. These problems include uncovering growth mechanisms driving the evolution of networks, predicting protein network fragility, and identifying environmental niches for metabolic networks. Deep learning offers a principled approach for mining complex networks and tackling graph-theoretic problems.

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15. Comparison of Different Generalizations of Clustering Coefficient and Local Efficiency for Weighted Undirected Graphs

Yu Wang, Eshwar Gorakhnath Ghumare, Rik Vandenberghe, Patrick Dupont

https://doi.org/10.1162/neco_a_00914 | pp. 313-331

Binary undirected graphs are well established, but when these graphs are constructed, often a threshold is applied to a parameter describing the connection between two nodes. Therefore, the use of weighted graphs is more appropriate. In this work, we focus on weighted undirected graphs. This implies that we have to incorporate edge weights in the graph measures, which require generalizations of common graph metrics. After reviewing existing generalizations of the clustering coefficient and the local efficiency, we proposed new generalizations for these graph measures. To be able to compare different generalizations, a number of essential and useful properties were defined that ideally should be satisfied. We applied the generalizations to two real-world networks of different sizes. As a result, we found that not all existing generalizations satisfy all essential properties. Furthermore, we determined the best generalization for the clustering coefficient and local efficiency based on their properties and the performance when applied to two networks. We found that the best generalization of the clustering coefficient is [Formula: see text], defined in Miyajima and Sakuragawa (2014), while the best generalization of the local efficiency is [Formula: see text], proposed in this letter. Depending on the application and the relative importance of sensitivity and robustness to noise, other generalizations may be selected on the basis of the properties investigated in this letter.

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16. Time Series Decomposition into Oscillation Components and Phase Estimation

Takeru Matsuda, Fumiyasu Komaki

https://doi.org/10.1162/neco_a_00916 | pp. 332-367

Many time series are naturally considered as a superposition of several oscillation components. For example, electroencephalogram (EEG) time series include oscillation components such as alpha, beta, and gamma. We propose a method for decomposing time series into such oscillation components using state-space models. Based on the concept of random frequency modulation, gaussian linear state-space models for oscillation components are developed. In this model, the frequency of an oscillator fluctuates by noise. Time series decomposition is accomplished by this model like the Bayesian seasonal adjustment method. Since the model parameters are estimated from data by the empirical Bayes' method, the amplitudes and the frequencies of oscillation components are determined in a data-driven manner. Also, the appropriate number of oscillation components is determined with the Akaike information criterion (AIC). In this way, the proposed method provides a natural decomposition of the given time series into oscillation components. In neuroscience, the phase of neural time series plays an important role in neural information processing. The proposed method can be used to estimate the phase of each oscillation component and has several advantages over a conventional method based on the Hilbert transform. Thus, the proposed method enables an investigation of the phase dynamics of time series. Numerical results show that the proposed method succeeds in extracting intermittent oscillations like ripples and detecting the phase reset phenomena. We apply the proposed method to real data from various fields such as astronomy, ecology, tidology, and neuroscience.

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17. Neural Circuits Trained with Standard Reinforcement Learning Can Accumulate Probabilistic Information during Decision Making

Nils Kurzawa, Christopher Summerfield, Rafa■ Bogacz

https://doi.org/10.1162/neco_a_00917 | pp. 368-393

Much experimental evidence suggests that during decision making, neural circuits accumulate evidence supporting alternative options. A computational model well describing this accumulation for choices between two options assumes that the brain integrates the log ratios of the likelihoods of the sensory inputs given the two options. Several models have been proposed for how neural circuits can learn these log-likelihood ratios from experience, but all of these models introduced novel and specially dedicated synaptic plasticity rules. Here we show that for a certain wide class of tasks, the log-likelihood ratios are approximately linearly proportional to the expected rewards for selecting actions. Therefore, a simple model based on standard reinforcement learning rules is able to estimate the log-likelihood ratios from experience and on each trial accumulate the log-likelihood ratios associated with presented stimuli while selecting an action. The simulations of the model replicate experimental data on both behavior and neural activity in tasks requiring accumulation of probabilistic cues. Our results suggest that there is no need for the brain to support dedicated plasticity rules, as the standard mechanisms proposed to describe reinforcement learning can enable the neural circuits to perform efficient probabilistic inference.

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18. Local and Global Gestalt Laws: A Neurally Based Spectral Approach

Marta Favali, Giovanna Citti, Alessandro Sarti

https://doi.org/10.1162/neco_a_00921 | pp. 394-422

This letter presents a mathematical model of figure-ground articulation that takes into account both local and global gestalt laws and is compatible with the functional architecture of the primary visual cortex (V1). The local gestalt law of good continuation is described by means of suitable connectivity kernels that are derived from Lie group theory and quantitatively compared with long-range connectivity in V1. Global gestalt constraints are then introduced in terms of spectral analysis of a connectivity matrix derived from these kernels. This analysis performs grouping of local features and individuates perceptual units with the highest salience. Numerical simulations are performed, and results are obtained by applying the technique to a number of stimuli.

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19. Multistability of Delayed Recurrent Neural Networks with Mexican Hat Activation Functions

Peng Liu, Zhigang Zeng, Jun Wang

https://doi.org/10.1162/neco_a_00922 | pp. 423-457

This letter studies the multistability analysis of delayed recurrent neural networks with Mexican hat activation function. Some sufficient conditions are obtained to ensure that an [Formula: see text]-dimensional recurrent neural network can have [Formula: see text] equilibrium points with [Formula: see text], and [Formula: see text] of them are locally exponentially stable. Furthermore, the attraction basins of these stable equilibrium points are estimated. We show that the attraction basins of these stable equilibrium points can be larger than their originally partitioned subsets. The results of this letter improve and extend the existing stability

results in the literature. Finally, a numerical example containing different cases is given to illustrate the theoretical results.

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20. Orthogonal Connectivity Factorization: Interpretable Decomposition of Variability in Correlation Matrices

Aapo Hyvärinen, Jun-ichiro Hirayama, Vesa Kiviniemi, Motoaki Kawanabe
https://doi.org/10.1162/neco_a_00810 | pp. 445-484

In many multivariate time series, the correlation structure is nonstationary, that is, it changes over time. The correlation structure may also change as a function of other cofactors, for example, the identity of the subject in biomedical data. A fundamental approach for the analysis of such data is to estimate the correlation structure (connectivities) separately in short time windows or for different subjects and use existing machine learning methods, such as principal component analysis (PCA), to summarize or visualize the changes in connectivity. However, the visualization of such a straightforward PCA is problematic because the ensuing connectivity patterns are much more complex objects than, say, spatial patterns. Here, we develop a new framework for analyzing variability in connectivities using the PCA approach as the starting point. First, we show how to analyze and visualize the principal components of connectivity matrices by a tailor-made rank-two matrix approximation in which we use the outer product of two orthogonal vectors. This leads to a new kind of transformation of eigenvectors that is particularly suited for this purpose and often enables interpretation of the principal component as connectivity between two groups of variables. Second, we show how to incorporate the orthogonality and the rank-two constraint in the estimation of PCA itself to improve the results. We further provide an interpretation of these methods in terms of estimation of a probabilistic generative model related to blind separation of dependent sources. Experiments on brain imaging data give very promising results.

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21. Exponentially Long Orbits in Hopfield Neural Networks

Samuel P. Muscinelli, Wulfram Gerstner, Johanni Brea
https://doi.org/10.1162/neco_a_00919 | pp. 458-484

We show that Hopfield neural networks with synchronous dynamics and asymmetric weights admit stable orbits that form sequences of maximal length. For [Formula: see text] units, these sequences have length [Formula: see text]; that is, they cover the full state space. We present a mathematical proof that maximal-length orbits exist for all [Formula: see text], and we provide a method to construct both the sequence and the weight matrix that allow its production. The orbit is relatively robust to dynamical noise, and perturbations of the optimal weights reveal other periodic orbits that are not maximal but typically still very long. We discuss how the resulting dynamics on slow time-scales can be used to generate desired output sequences.

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22. A Block Successive Lower-Bound Maximization Algorithm for the Maximum Pseudo-Likelihood Estimation of Fully Visible Boltzmann Machines

Hien D. Nguyen, Ian Wood

https://doi.org/10.1162/neco_a_00813 | pp. 485-492

Maximum pseudo-likelihood estimation (MPLE) is an attractive method for training fully visible Boltzmann machines (FVBMs) due to its computational scalability and the desirable statistical properties of the MPLE. No published algorithms for MPLE have been proven to be convergent or monotonic. In this note, we present an algorithm for the MPLE of FVBMs based on the block successive lower-bound maximization (BSLM) principle. We show that the BSLM algorithm monotonically increases the pseudo-likelihood values and that the sequence of BSLM estimates converges to the unique global maximizer of the pseudo-likelihood function. The relationship between the BSLM algorithm and the gradient ascent (GA) algorithm for MPLE of FVBMs is also discussed, and a convergence criterion for the GA algorithm is given.

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23. Controllability Analysis of the Neural Mass Model with Dynamic Parameters

Xian Liu, Jing Gao, Guan Wang, Chen Zhi-wang

https://doi.org/10.1162/neco_a_00925 | pp. 485-501

The development of control technology for the brain is of potential significance to the prevention and treatment of neuropsychiatric disorders and the improvement of humans' mental health. A controllability analysis of the brain is necessary to ensure the feasibility of the brain control. In this letter, we investigate the influences of dynamical parameters on the controllability in the neural mass model by using controllability indices as quantitative indicators. The indices are obtained by computing Lie brackets and condition numbers of the system model. We show how controllability changes with important parameters of our dynamical (neuronal) model. Our results suggest that the underlying dynamical parameters have certain ranges with better controllability. We hope it can play potential roles in therapy for brain nervous disorder disease.

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24. Microdomain [Ca²⁺] Fluctuations Alter Temporal Dynamics in Models of Ca²⁺-Dependent Signaling Cascades and Synaptic Vesicle Release

Seth H. Weinberg

https://doi.org/10.1162/neco_a_00811 | pp. 493-524

Ca(2+)-dependent signaling is often localized in spatially restricted microdomains and may involve only 1 to 100 Ca(2+) ions. Fluctuations in the microdomain Ca(2+) concentration (Ca(2+)) can arise from a wide range of elementary processes, including diffusion, Ca(2+) influx, and association/dissociation with Ca(2+) binding proteins or buffers. However, it is unclear to what extent these fluctuations alter Ca(2+)-dependent signaling. We construct Markov models of a general Ca(2+)-dependent signaling cascade and Ca(2+)-triggered synaptic vesicle release. We compare the hitting (release) time distribution and statistics for models that account for [Ca(2+)] fluctuations with the corresponding models that neglect these fluctuations. In general, when Ca(2+) fluctuations are much faster than the characteristic time for the signaling event, the hitting time distributions and statistics for the models with and without Ca(2+) fluctuation are similar. However, when the timescale of Ca(2+) fluctuations is on the same order as the signaling cascade or slower, the hitting time mean and variability are typically increased, in particular when the

average number of microdomain Ca(2+) ions is small, a consequence of a long-tailed hitting time distribution. In a model of Ca(2+)-triggered synaptic vesicle release, we demonstrate the conditions for which [Ca(2+)] fluctuations do and do not alter the distribution, mean, and variability of release timing. We find that both the release time mean and variability can be increased, demonstrating that Ca(2+) fluctuations are an important aspect of microdomain Ca(2+) signaling and further suggesting that Ca(2+) fluctuations in the presynaptic terminal may contribute to variability in synaptic vesicle release and thus variability in neuronal spiking.

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25. Energy Model of Neuron Activation

Yuriy Romanyshyn, Andriy Smerdov, Svitlana Petrytska

https://doi.org/10.1162/neco_a_00913 | pp. 502-518

On the basis of the neurophysiological strength-duration (amplitude-duration) curve of neuron activation (which relates the threshold amplitude of a rectangular current pulse of neuron activation to the pulse duration), as well as with the use of activation energy constraint (the threshold curve corresponds to the energy threshold of neuron activation by a rectangular current pulse), an energy model of neuron activation by a single current pulse has been constructed. The constructed model of activation, which determines its spectral properties, is a bandpass filter. Under the condition of minimum-phase feature of the neuron activation model, on the basis of Hilbert transform, the possibilities of phase-frequency response calculation from its amplitude-frequency response have been considered. Approximation to the amplitude-frequency response by the response of the Butterworth filter of the first order, as well as obtaining the pulse response corresponding to this approximation, give us the possibility of analyzing the efficiency of activating current pulses of various shapes, including analysis in accordance with the energy constraint.

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26. Block-Regularizedm×2Cross-Validated Estimator of the Generalization Error

Ruibo Wang, Yu Wang, Jihong Li, Xingli Yang, Jing Yang

https://doi.org/10.1162/neco_a_00923 | pp. 519-554

A cross-validation method based on [Formula: see text] replications of two-fold cross validation is called an [Formula: see text] cross validation. An [Formula: see text] cross validation is used in estimating the generalization error and comparing of algorithms' performance in machine learning. However, the variance of the estimator of the generalization error in [Formula: see text] cross validation is easily affected by random partitions. Poor data partitioning may cause a large fluctuation in the number of overlapping samples between any two training (test) sets in [Formula: see text] cross validation. This fluctuation results in a large variance in the [Formula: see text] cross-validated estimator. The influence of the random partitions on variance becomes serious as [Formula: see text] increases. Thus, in this study, the partitions with a restricted number of overlapping samples between any two training (test) sets are defined as a block-regularized partition set. The corresponding cross validation is called block-regularized [Formula: see text] cross validation ([Formula: see text] BCV). It can effectively reduce the influence of random partitions. We prove that the variance of the [Formula: see text] BCV estimator of the generalization error is smaller than the variance of [Formula: see text] cross-validated estimator and reaches the minimum in a special situation. An analytical expression of the variance can also be derived in this special situation. This

conclusion is validated through simulation experiments. Furthermore, a practical construction method of [Formula: see text] BCV by a two-level orthogonal array is provided. Finally, a conservative estimator is proposed for the variance of estimator of the generalization error.

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27. Kernelized Elastic Net Regularization: Generalization Bounds, and Sparse Recovery

Yunlong Feng, Shaogao Lv, Hanyuan Hang, Johan A. K. Suykens

https://doi.org/10.1162/neco_a_00812 | pp. 525-562

Kernelized elastic net regularization (KENReg) is a kernelization of the well-known elastic net regularization (Zou & Hastie, 2005). The kernel in KENReg is not required to be a Mercer kernel since it learns from a kernelized dictionary in the coefficient space. Feng, Yang, Zhao, Lv, and Suykens (2014) showed that KENReg has some nice properties including stability, sparseness, and generalization. In this letter, we continue our study on KENReg by conducting a refined learning theory analysis. This letter makes the following three main contributions. First, we present refined error analysis on the generalization performance of KENReg. The main difficulty of analyzing the generalization error of KENReg lies in characterizing the population version of its empirical target function. We overcome this by introducing a weighted Banach space associated with the elastic net regularization. We are then able to conduct elaborated learning theory analysis and obtain fast convergence rates under proper complexity and regularity assumptions. Second, we study the sparse recovery problem in KENReg with fixed design and show that the kernelization may improve the sparse recovery ability compared to the classical elastic net regularization. Finally, we discuss the interplay among different properties of KENReg that include sparseness, stability, and generalization. We show that the stability of KENReg leads to generalization, and its sparseness confidence can be derived from generalization. Moreover, KENReg is stable and can be simultaneously sparse, which makes it attractive theoretically and practically.

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28. An Online Policy Gradient Algorithm for Markov Decision Processes with Continuous States and Actions

Yao Ma, Tingting Zhao, Kohei Hatano, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00808 | pp. 563-593

We consider the learning problem under an online Markov decision process (MDP) aimed at learning the time-dependent decision-making policy of an agent that minimizes the regret-the difference from the best fixed policy. The difficulty of online MDP learning is that the reward function changes over time. In this letter, we show that a simple online policy gradient algorithm achieves regret $O(\sqrt{T})$ for T steps under a certain concavity assumption and $O(\log T)$ under a strong concavity assumption. To the best of our knowledge, this is the first work to present an online MDP algorithm that can handle continuous state, action, and parameter spaces with guarantee. We also illustrate the behavior of the proposed online policy gradient method through experiments.

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29. The Development of an Information Criterion for Change-Point Analysis

Colin LaMont, Paul A. Wiggins
https://doi.org/10.1162/neco_a_00809 | pp. 594-612

Change-point analysis is a flexible and computationally tractable tool for the analysis of times series data from systems that transition between discrete states and whose observables are corrupted by noise. The change point algorithm is used to identify the time indices (change points) at which the system transitions between these discrete states. We present a unified information-based approach to testing for the existence of change points. This new approach reconciles two previously disparate approaches to change-point analysis (frequentist and information based) for testing transitions between states. The resulting method is statistically principled, parameter and prior free, and widely applicable to a wide range of change-point problems.

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30. Vector Symbolic Spiking Neural Network Model of Hippocampal Subarea CA1 Novelty Detection Functionality

Claus Agerskov
https://doi.org/10.1162/neco_a_00826 | pp. 613-628

A neural network model is presented of novelty detection in the CA1 subdomain of the hippocampal formation from the perspective of information flow. This computational model is restricted on several levels by both anatomical information about hippocampal circuitry and behavioral data from studies done in rats. Several studies report that the CA1 area broadcasts a generalized novelty signal in response to changes in the environment. Using the neural engineering framework developed by Eliasmith et al., a spiking neural network architecture is created that is able to compare high-dimensional vectors, symbolizing semantic information, according to the semantic pointer hypothesis. This model then computes the similarity between the vectors, as both direct inputs and a recalled memory from a long-term memory network by performing the dot-product operation in a novelty neural network architecture. The developed CA1 model agrees with available neuroanatomical data, as well as the presented behavioral data, and so it is a biologically realistic model of novelty detection in the hippocampus, which can provide a feasible explanation for experimentally observed dynamics.

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31. Feature-Based Attention by Lateral Spike Synchronization

August Romeo, Hans Supèr
https://doi.org/10.1162/neco_a_00822 | pp. 629-651

We introduce a neural model capable of feature selectiveness by spike-mediated synchronization through lateral synaptic couplings. For a stimulus containing two features, the attended one elicits a higher response. In the case of sequential single-feature stimuli, repetition of the attended feature also results in an enhanced response, exhibited by greater synchrony and higher spiking rates.

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32. Selective Interareal Synchronization through Gamma Frequency Differences and Slower-Rhythm Gamma Phase Reset

Thomas Burwick, Alexandros Bouras

https://doi.org/10.1162/neco_a_00908 | pp. 643-678

The communication-through-coherence (CTC) hypothesis states that a sending group of neurons will have a particularly strong effect on a receiving group if both groups oscillate in a phase-locked ("coherent") manner (Fries, 2005 , 2015). Here, we consider a situation with two visual stimuli, one in the focus of attention and the other distracting, resulting in two sites of excitation at an early cortical area that project to a common site in a next area. Taking a modeler's perspective, we confirm the workings of a mechanism that was proposed by Bosman et al. (2012) in the context of providing experimental evidence for the CTC hypothesis: a slightly higher gamma frequency of the attended sending site compared to the distracting site may cause selective interareal synchronization with the receiving site if combined with a slow-rhythm gamma phase reset. We also demonstrate the relevance of a slightly lower intrinsic frequency of the receiving site for this scenario. Moreover, we discuss conditions for a transition from bottom-up to top-down driven phase locking.

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33. Downstream Effect of Ramping Neuronal Activity through Synapses with Short-Term Plasticity

Wei Wei, Xiao■Jing Wang

https://doi.org/10.1162/neco_a_00818 | pp. 652-666

Ramping neuronal activity refers to spiking activity with a rate that increases quasi-linearly over time. It has been observed in multiple cortical areas and is correlated with evidence accumulation processes or timing. In this work, we investigated the downstream effect of ramping neuronal activity through synapses that display short-term facilitation (STF) or depression (STD). We obtained an analytical result for a synapse driven by deterministic linear ramping input that exhibits pure STF or STD and numerically investigated the general case when a synapse displays both STF and STD. We show that the analytical deterministic solution gives an accurate description of the averaging synaptic activation of many inputs converging onto a postsynaptic neuron, even when fluctuations in the ramping input are strong. Activation of a synapse with STF shows an initial cubical increase with time, followed by a linear ramping similar to a synapse without STF. Activation of a synapse with STD grows in time to a maximum before falling and reaching a plateau, and this steady state is independent of the slope of the ramping input. For a synapse displaying both STF and STD, an increase in the depression time constant from a value much smaller than the facilitation time constant $\tau(F)$ to a value much larger than $\tau(F)$ leads to a transition from facilitation dominance to depression dominance. Therefore, our work provides insights into the impact of ramping neuronal activity on downstream neurons through synapses that display short-term plasticity. In a perceptual decision-making process, ramping activity has been observed in the parietal and prefrontal cortices, with a slope that decreases with task difficulty. Our work predicts that neurons downstream from such a decision circuit could instead display a firing plateau independent of the task difficulty, provided that the synaptic connection is endowed with short-term depression.

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34. Optimal Schedules in Multitask Motor Learning

Jeong-Yoon Lee, Young-Min Oh, Sungshin Kim, Robert A. Scheidt, Nicolas Schweighofer

https://doi.org/10.1162/neco_a_00823 | pp. 667-685

Although scheduling multiple tasks in motor learning to maximize long-term retention of performance is of great practical importance in sports training and motor rehabilitation after brain injury, it is unclear how to do so. We propose here a novel theoretical approach that uses optimal control theory and computational models of motor adaptation to determine schedules that maximize long-term retention predictively. Using Pontryagin's maximum principle, we derived a control law that determines the trial-by-trial task choice that maximizes overall delayed retention for all tasks, as predicted by the state-space model. Simulations of a single session of adaptation with two tasks show that when task interference is high, there exists a threshold in relative task difficulty below which the alternating schedule is optimal. Only for large differences in task difficulties do optimal schedules assign more trials to the harder task. However, over the parameter range tested, alternating schedules yield long-term retention performance that is only slightly inferior to performance given by the true optimal schedules. Our results thus predict that in a large number of learning situations wherein tasks interfere, intermixing tasks with an equal number of trials is an effective strategy in enhancing long-term retention.

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35. Theoretical and Experimental Analyses of Tensor-Based Regression and Classification

Kishan Wimalawarne, Ryota Tomioka, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00815 | pp. 686-715

We theoretically and experimentally investigate tensor-based regression and classification. Our focus is regularization with various tensor norms, including the overlapped trace norm, the latent trace norm, and the scaled latent trace norm. We first give dual optimization methods using the alternating direction method of multipliers, which is computationally efficient when the number of training samples is moderate. We then theoretically derive an excess risk bound for each tensor norm and clarify their behavior. Finally, we perform extensive experiments using simulated and real data and demonstrate the superiority of tensor-based learning methods over vector- and matrix-based learning methods.

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36. Feature Selection for Ridge Regression with Provable Guarantees

Saurabh Paul, Petros Drineas

https://doi.org/10.1162/neco_a_00816 | pp. 716-742

We introduce single-set spectral sparsification as a deterministic sampling-based feature selection technique for regularized least-squares classification, which is the classification analog to ridge regression. The method is unsupervised and gives worst-case guarantees of the generalization power of the classification function after feature selection with respect to the classification function obtained using all features. We also introduce leverage-score sampling as an unsupervised randomized feature selection method for ridge regression. We provide risk bounds for both single-set spectral sparsification and leverage-score sampling on ridge regression in the fixed design setting and show that the risk in the sampled space is comparable to the risk in the full-feature space. We perform experiments on synthetic and real-world data sets; a subset of TechTC-300 data sets, to support our theory. Experimental results indicate that the proposed methods perform better than the existing feature selection methods.

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37. On Decoding Grid Cell Population Codes Using Approximate Belief Propagation

Yongseok Yoo, Woori Kim

https://doi.org/10.1162/neco_a_00902 | pp. 716-734

Neural systems are inherently noisy. One well-studied example of a noise reduction mechanism in the brain is the population code, where representing a variable with multiple neurons allows the encoded variable to be recovered with fewer errors. Studies have assumed ideal observer models for decoding population codes, and the manner in which information in the neural population can be retrieved remains elusive. This letter addresses a mechanism by which realistic neural circuits can recover encoded variables. Specifically, the decoding problem of recovering a spatial location from populations of grid cells is studied using belief propagation. We extend the belief propagation decoding algorithm in two aspects. First, beliefs are approximated rather than being calculated exactly. Second, decoding noises are introduced into the decoding circuits. Numerical simulations demonstrate that beliefs can be effectively approximated by combining polynomial nonlinearities with divisive normalization. This approximate belief propagation algorithm is tolerant to decoding noises. Thus, this letter presents a realistic model for decoding neural population codes and investigates fault-tolerant information retrieval mechanisms in the brain.

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38. Online Pairwise Learning Algorithms

Yiming Ying, Ding■Xuan Zhou

https://doi.org/10.1162/neco_a_00817 | pp. 743-777

Pairwise learning usually refers to a learning task that involves a loss function depending on pairs of examples, among which the most notable ones are bipartite ranking, metric learning, and AUC maximization. In this letter we study an online algorithm for pairwise learning with a least-square loss function in an unconstrained setting of a reproducing kernel Hilbert space (RKHS) that we refer to as the Online Pairwise lEaRning Algorithm (OPERA). In contrast to existing works (Kar, Sriperumbudur, Jain, & Karnick, 2013 ; Wang, Khardon, Pechyony, & Jones, 2012), which require that the iterates are restricted to a bounded domain or the loss function is strongly convex, OPERA is associated with a non-strongly convex objective function and learns the target function in an unconstrained RKHS. Specifically, we establish a general theorem that guarantees the almost sure convergence for the last iterate of OPERA without any assumptions on the underlying distribution. Explicit convergence rates are derived under the condition of polynomially decaying step sizes. We also establish an interesting property for a family of widely used kernels in the setting of pairwise learning and illustrate the convergence results using such kernels. Our methodology mainly depends on the characterization of RKHSs using its associated integral operators and probability inequalities for random variables with values in a Hilbert space.

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39. Sampled-Data State Feedback Stabilization of Boolean Control Networks

Yang Liu, Jinde Cao, Liangjie Sun, Jianquan Lu

https://doi.org/10.1162/neco_a_00819 | pp. 778-799

In this letter, we investigate the sampled-data state feedback control (SDSFC) problem of Boolean control networks (BCNs). Some necessary and sufficient conditions are obtained for the global stabilization of BCNs

by SDSFC. Different from conventional state feedback controls, new phenomena observed the study of SDSFC. Based on the controllability matrix, we derive some necessary and sufficient conditions under which the trajectories of BCNs can be stabilized to a fixed point by piecewise constant control (PCC). It is proved that the global stabilization of BCNs under SDSFC is equivalent to that by PCC. Moreover, algorithms are given to construct the sampled-data state feedback controllers. Numerical examples are given to illustrate the efficiency of the obtained results.

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40. Causal Inference on Discrete Data via Estimating Distance Correlations

Furui Liu, Laiwan Chan

https://doi.org/10.1162/neco_a_00820 | pp. 801-814

In this article, we deal with the problem of inferring causal directions when the data are on discrete domain. By considering the distribution of the cause [Formula: see text] and the conditional distribution mapping cause to effect [Formula: see text] as independent random variables, we propose to infer the causal direction by comparing the distance correlation between [Formula: see text] and [Formula: see text] with the distance correlation between [Formula: see text] and [Formula: see text]. We infer that X causes Y if the dependence coefficient between [Formula: see text] and [Formula: see text] is smaller. Experiments are performed to show the performance of the proposed method.

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41. Multiple Diffusion Models to Compare Saccadic and Manual Responses for Inhibition of Return

W. Joseph MacInnes

https://doi.org/10.1162/neco_a_00904 | pp. 804-824

Cuing a location in space produces a short-lived advantage in reaction time to targets at that location. This early advantage, however, switches to a reaction time cost and has been termed inhibition of return (IOR). IOR behaves differently for different response modalities, suggesting that it may not be a unified effect. This letter presents new data from two experiments testing the gradient of IOR with random, continuous cue-target Euclidean distance and cue-target onset asynchrony. These data were then used to train multiple diffusion models of saccadic and manual reaction time for these cuing experiments. Diffusion models can generate accurate distributions of reaction time data by modeling a response as a buildup of evidence toward a response threshold. If saccadic and attentional IOR are based on similar processes, then differences in distribution will be best explained by adjusting parameter values such as signal and noise within the same model structure. Although experimental data show differences in the timing of IOR across modality, best-fit models are shown to have similar model parameters for the gradient of IOR, suggesting similar underlying mechanisms for saccadic and manual IOR.

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42. A Mathematical Motivation for Complex-Valued Convolutional Networks

Mark Tygert, Joan Bruna, Soumith Chintala, Yann LeCun, Serkan Piantino, Arthur Szlam

https://doi.org/10.1162/neco_a_00824 | pp. 815-825

A complex-valued convolutional network (convnet) implements the repeated application of the following composition of three operations, recursively applying the composition to an input vector of nonnegative real numbers: (1) convolution with complex-valued vectors, followed by (2) taking the absolute value of every entry of the resulting vectors, followed by (3) local averaging. For processing real-valued random vectors, complex-valued convnets can be viewed as data-driven multiscale windowed power spectra, data-driven multiscale windowed absolute spectra, data-driven multiwavelet absolute values, or (in their most general configuration) data-driven nonlinear multiwavelet packets. Indeed, complex-valued convnets can calculate multiscale windowed spectra when the convnet filters are windowed complex-valued exponentials. Standard real-valued convnets, using rectified linear units (ReLUs), sigmoidal (e.g., logistic or tanh) nonlinearities, or max pooling, for example, do not obviously exhibit the same exact correspondence with data-driven wavelets (whereas for complex-valued convnets, the correspondence is much more than just a vague analogy). Courtesy of the exact correspondence, the remarkably rich and rigorous body of mathematical analysis for wavelets applies directly to (complex-valued) convnets.

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43. Learning Precise Spike Train–to–Spike Train Transformations in Multilayer Feedforward Neuronal Networks

Arunava Banerjee

https://doi.org/10.1162/neco_a_00829 | pp. 826-848

We derive a synaptic weight update rule for learning temporally precise spike train-to-spike train transformations in multilayer feedforward networks of spiking neurons. The framework, aimed at seamlessly generalizing error backpropagation to the deterministic spiking neuron setting, is based strictly on spike timing and avoids invoking concepts pertaining to spike rates or probabilistic models of spiking. The derivation is founded on two innovations. First, an error functional is proposed that compares the spike train emitted by the output neuron of the network to the desired spike train by way of their putative impact on a virtual postsynaptic neuron. This formulation sidesteps the need for spike alignment and leads to closed-form solutions for all quantities of interest. Second, virtual assignment of weights to spikes rather than synapses enables a perturbation analysis of individual spike times and synaptic weights of the output, as well as all intermediate neurons in the network, which yields the gradients of the error functional with respect to the said entities. Learning proceeds via a gradient descent mechanism that leverages these quantities. Simulation experiments demonstrate the efficacy of the proposed learning framework. The experiments also highlight asymmetries between synapses on excitatory and inhibitory neurons.

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44. Separating Spike Count Correlation from Firing Rate Correlation

Giuseppe Vinci, Valérie Ventura, Matthew A. Smith, Robert E. Kass

https://doi.org/10.1162/neco_a_00831 | pp. 849-881

Populations of cortical neurons exhibit shared fluctuations in spiking activity over time. When measured for a pair of neurons over multiple repetitions of an identical stimulus, this phenomenon emerges as correlated trial-to-trial response variability via spike count correlation (SCC). However, spike counts can be viewed as noisy versions of firing rates, which can vary from trial to trial. From this perspective, the SCC for a pair of neurons becomes a noisy version of the corresponding firing rate correlation (FRC). Furthermore, the magnitude of the SCC is generally smaller than that of the FRC and is likely to be less sensitive to

experimental manipulation. We provide statistical methods for disambiguating time-averaged drive from within-trial noise, thereby separating FRC from SCC. We study these methods to document their reliability, and we apply them to neurons recorded in vivo from area V4 in an alert animal. We show how the various effects we describe are reflected in the data: within-trial effects are largely negligible, while attenuation due to trial-to-trial variation dominates and frequently produces comparisons in SCC that, because of noise, do not accurately reflect those based on the underlying FRC.

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45. Effect of Reference Scheme on Power and Phase of the Local Field Potential

Vinay Shirhatti, Ayon Borthakur, Supratim Ray

https://doi.org/10.1162/neco_a_00827 | pp. 882-913

Brain signals are often analyzed in the spectral domain, where the power spectral density (PSD) and phase differences and consistency can reveal important information about the network. However, for proper interpretation, it is important to know whether these measures depend on stimulus/behavioral conditions or the reference scheme used to analyze data. We recorded local field potential (LFP) from an array of microelectrodes chronically implanted in area V1 of monkeys under different stimulus/behavioral conditions and computed PSD slopes, coherence, and phase difference between LFPs as a function of frequency and interelectrode distance while using four reference schemes: single wire, average, bipolar, and current source density. PSD slopes were dependent on reference scheme at low frequencies (below 200 Hz) but became invariant at higher frequencies. Average phase differences between sites also depended critically on referencing, switching from 0 degrees for single-wire to 180 degrees for average reference. Results were consistent across different stimulus/behavioral conditions. We were able to account for these results based on the coherence profile across sites and properties of the spectral estimator. Our results show that using different reference schemes can have drastic effects on phase differences and PSD slopes and therefore must be interpreted carefully to gain insights about network properties.

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46. State-Space Analysis of Granger-Geweke Causality Measures with Application to fMRI

Victor Solo

https://doi.org/10.1162/neco_a_00828 | pp. 914-949

The recent interest in the dynamics of networks and the advent, across a range of applications, of measuring modalities that operate on different temporal scales have put the spotlight on some significant gaps in the theory of multivariate time series. Fundamental to the description of network dynamics is the direction of interaction between nodes, accompanied by a measure of the strength of such interactions. Granger causality and its associated frequency domain strength measures (GEMs) (due to Geweke) provide a framework for the formulation and analysis of these issues. In pursuing this setup, three significant unresolved issues emerge. First, computing GEMs involves computing submodels of vector time series models, for which reliable methods do not exist. Second, the impact of filtering on GEMs has never been definitively established. Third, the impact of downsampling on GEMs has never been established. In this work, using state-space methods, we resolve all these issues and illustrate the results with some simulations. Our analysis is motivated by some problems in (fMRI) brain imaging, to which we apply it, but it

is of general applicability.

47. Whole-Body Reaching Movements Formulated by Minimum Muscle-Tension Change Criterion

Naoki Kudo, Kyuheong Choi, Takahiro Kagawa, Yoji Uno

https://doi.org/10.1162/neco_a_00830 | pp. 950-969

It is well known that planar reaching movements of the human shoulder and elbow joints have invariant features: roughly straight hand paths and bell-shaped velocity profiles. The optimal control models with the criteria of smoothness or precision, which determine a unique movement pattern, predict such features of hand trajectories. In this letter on expanding the research on simple arm reaching movements, we examine whether the smoothness criteria can be applied to whole-body reaching movements with many degrees of freedom. Determining a suitable joint trajectory in the whole-body reaching movement corresponds to the optimization problem with constraints, since body balance must be maintained during a motion task. First, we measured human joint trajectories and ground reaction forces during whole-body reaching movements, and confirmed that subjects formed similar movements with common characteristics in the trajectories of the hand position and body center of mass. Second, we calculated the optimal trajectories according to the criteria of torque and muscle-tension smoothness. While the minimum torque change trajectories were not consistent with the experimental data, the minimum muscle-tension change model was able to predict the stereotyped features of the measured trajectories. To explore the dominant effects of the extension from the torque change to the muscle-tension change, we introduced a weighted torque change cost function. Considering the maximum voluntary contraction (MVC) force of the muscle as the weighting factor of each joint torque, we formulated the weighted torque change cost as a simplified version of the minimum muscle-tension change cost. The trajectories owing to the minimum weighted torque change criterion also showed qualitative agreement with the common features of the measured data. Proper estimation of the MVC forces in the body joints is essential to reproduce human whole-body movements according to the minimum muscle-tension change criterion.

48. Recurrent Neural Network for Computing Outer Inverse

Ivan S. Živkovi■, Predrag S. Stanimirovi■, Yimin Wei

https://doi.org/10.1162/neco_a_00821 | pp. 970-998

Two linear recurrent neural networks for generating outer inverses with prescribed range and null space are defined. Each of the proposed recurrent neural networks is based on the matrix-valued differential equation, a generalization of dynamic equations proposed earlier for the nonsingular matrix inversion, the Moore-Penrose inversion, as well as the Drazin inversion, under the condition of zero initial state. The application of the first approach is conditioned by the properties of the spectrum of a certain matrix; the second approach eliminates this drawback, though at the cost of increasing the number of matrix operations. The cases corresponding to the most common generalized inverses are defined. The conditions that ensure stability of the proposed neural network are presented. Illustrative examples present the results of numerical simulations.

49. Electroencephalographic Motor Imagery Brain Connectivity Analysis for BCI: A Review

Mahyar Hamedi, Sh-Hussain Salleh, Alias Mohd Noor

https://doi.org/10.1162/neco_a_00838 | pp. 999-1041

Recent research has reached a consensus on the feasibility of motor imagery brain-computer interface (MI-BCI) for different applications, especially in stroke rehabilitation. Most MI-BCI systems rely on temporal, spectral, and spatial features of single channels to distinguish different MI patterns. However, no successful communication has been established for a completely locked-in subject. To provide more useful and informative features, it has been recommended to take into account the relationships among electroencephalographic (EEG) sensor/source signals in the form of brain connectivity as an efficient tool of neuroscience. In this review, we briefly report the challenges and limitations of conventional MI-BCIs. Brain connectivity analysis, particularly functional and effective, has been described as one of the most promising approaches for improving MI-BCI performance. An extensive literature on EEG-based MI brain connectivity analysis of healthy subjects is reviewed. We subsequently discuss the brain connectomes during left and right hand, feet, and tongue MI movements. Moreover, key components involved in brain connectivity analysis that considerably affect the results are explained. Finally, possible technical shortcomings that may have influenced the results in previous research are addressed and suggestions are provided.

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50. McCulloch-Pitts Brains and Pseudorandom Functions

Vašek Chvátal, M. Goldsmith, Nan Yang

https://doi.org/10.1162/neco_a_00841 | pp. 1042-1050

In a pioneering classic, Warren McCulloch and Walter Pitts proposed a model of the central nervous system. Motivated by EEG recordings of normal brain activity, Chvátal and Goldsmith asked whether these dynamical systems can be engineered to produce trajectories that are irregular, disorderly, and apparently unpredictable. We show that they cannot build weak pseudorandom functions.

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51. Maintaining Consistency of Spatial Information in the Hippocampal Network: A Combinatorial Geometry Model

Yuri Dabaghian

https://doi.org/10.1162/neco_a_00840 | pp. 1051-1071

Place cells in the rat hippocampus play a key role in creating the animal's internal representation of the world. During active navigation, these cells spike only in discrete locations, together encoding a map of the environment. Electrophysiological recordings have shown that the animal can revisit this map mentally during both sleep and awake states, reactivating the place cells that fired during its exploration in the same sequence in which they were originally activated. Although consistency of place cell activity during active navigation is arguably enforced by sensory and proprioceptive inputs, it remains unclear how a consistent representation of space can be maintained during spontaneous replay. We propose a model that can account for this phenomenon and suggest that a spatially consistent replay requires a number of constraints on the hippocampal network that affect its synaptic architecture and the statistics of synaptic connection strengths.

52. Feature-Linking Model for Image Enhancement

Kun Zhan, Jicai Teng, Jinhui Shi, Qiaoqiao Li, Mingying Wang

https://doi.org/10.1162/neco_a_00832 | pp. 1072-1100

Inspired by gamma-band oscillations and other neurobiological discoveries, neural networks research shifts the emphasis toward temporal coding, which uses explicit times at which spikes occur as an essential dimension in neural representations. We present a feature-linking model (FLM) that uses the timing of spikes to encode information. The first spiking time of FLM is applied to image enhancement, and the processing mechanisms are consistent with the human visual system. The enhancement algorithm achieves boosting the details while preserving the information of the input image. Experiments are conducted to demonstrate the effectiveness of the proposed method. Results show that the proposed method is effective.

53. Direct Density Derivative Estimation

Hiroaki Sasaki, Yung■Kyun Noh, Gang Niu, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00835 | pp. 1101-1140

Estimating the derivatives of probability density functions is an essential step in statistical data analysis. A naive approach to estimate the derivatives is to first perform density estimation and then compute its derivatives. However, this approach can be unreliable because a good density estimator does not necessarily mean a good density derivative estimator. To cope with this problem, in this letter, we propose a novel method that directly estimates density derivatives without going through density estimation. The proposed method provides computationally efficient estimation for the derivatives of any order on multidimensional data with a hyperparameter tuning method and achieves the optimal parametric convergence rate. We further discuss an extension of the proposed method by applying regularized multitask learning and a general framework for density derivative estimation based on Bregman divergences. Applications of the proposed method to nonparametric Kullback-Leibler divergence approximation and bandwidth matrix selection in kernel density estimation are also explored.

54. Robust Clustering Method in the Presence of Scattered Observations

Akifumi Notsu, Shinto Eguchi

https://doi.org/10.1162/neco_a_00833 | pp. 1141-1162

Contamination of scattered observations, which are either featureless or unlike the other observations, frequently degrades the performance of standard methods such as K-means and model-based clustering. In this letter, we propose a robust clustering method in the presence of scattered observations called Gamma-clust. Gamma-clust is based on a robust estimation for cluster centers using gamma-divergence. It provides a proper solution for clustering in which the distributions for clustered data are nonnormal, such as t-distributions with different variance-covariance matrices and degrees of freedom. As demonstrated in a simulation study and data analysis, Gamma-clust is more flexible and provides superior results compared to the robustified K-means and model-based clustering.

55. Efficient Nonnegative Matrix Factorization by DC Programming and DCA

Hoai An Le Thi, Xuan Thanh Vo, Tao Pham Dinh

https://doi.org/10.1162/neco_a_00836 | pp. 1163-1216

In this letter, we consider the nonnegative matrix factorization (NMF) problem and several NMF variants. Two approaches based on DC (difference of convex functions) programming and DCA (DC algorithm) are developed. The first approach follows the alternating framework that requires solving, at each iteration, two nonnegativity-constrained least squares subproblems for which DCA-based schemes are investigated. The convergence property of the proposed algorithm is carefully studied. We show that with suitable DC decompositions, our algorithm generates most of the standard methods for the NMF problem. The second approach directly applies DCA on the whole NMF problem. Two algorithms-one computing all variables and one deploying a variable selection strategy-are proposed. The proposed methods are then adapted to solve various NMF variants, including the nonnegative factorization, the smooth regularization NMF, the sparse regularization NMF, the multilayer NMF, the convex/convex-hull NMF, and the symmetric NMF. We also show that our algorithms include several existing methods for these NMF variants as special versions. The efficiency of the proposed approaches is empirically demonstrated on both real-world and synthetic data sets. It turns out that our algorithms compete favorably with five state-of-the-art alternating nonnegative least squares algorithms.

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56. Robust Support Vector Machines for Classification with Nonconvex and Smooth Losses

Yunlong Feng, Yuning Yang, Xiaolin Huang, Siamak Mehrkanoon, Johan A. K. Suykens

https://doi.org/10.1162/neco_a_00837 | pp. 1217-1247

This letter addresses the robustness problem when learning a large margin classifier in the presence of label noise. In our study, we achieve this purpose by proposing robustified large margin support vector machines. The robustness of the proposed robust support vector classifiers (RSVC), which is interpreted from a weighted viewpoint in this work, is due to the use of nonconvex classification losses. Besides the robustness, we also show that the proposed RSCV is simultaneously smooth, which again benefits from using smooth classification losses. The idea of proposing RSVC comes from M-estimation in statistics since the proposed robust and smooth classification losses can be taken as one-sided cost functions in robust statistics. Its Fisher consistency property and generalization ability are also investigated. Besides the robustness and smoothness, another nice property of RSVC lies in the fact that its solution can be obtained by solving weighted squared hinge loss-based support vector machine problems iteratively. We further show that in each iteration, it is a quadratic programming problem in its dual space and can be solved by using state-of-the-art methods. We thus propose an iteratively reweighted type algorithm and provide a constructive proof of its convergence to a stationary point. Effectiveness of the proposed classifiers is verified on both artificial and real data sets.

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57. Erratum

(authors unknown)

https://doi.org/10.1162/neco_e_00861 | pp. 1248-1248

In the May 2016 issue of Neural Computation (Volume 28, Number 5, pp. 914–949), the letter “State-Space Analysis of Granger-Geweke Causality Measures with Application to fMRI” by Victor Solo was printed with incomplete acknowledgments.The published acknowledgment reads:This work was partially supported by NIH grant P41EB015896.The complete acknowledgment should read:This work was partially supported by the NIH grant P41EB015896. The author would first like to thank Martinos Center colleagues Dr. Bruce Rosen and Dr. Robert Weiskoff who raised crucial questions during a joint discussion of Granger causality and fMRI (held in 1998), which ultimately led to this paper. Thanks are also due to Dr. Graham Goodwin who asked a critical question during an IEEE Control Engineering conference presentation that led to the work on the reverse problem. Special thanks are due to a referee who posed a series of critical questions that led to a major expansion of section 9 into its current form. Finally we acknowledge discussions with Martinos Center colleagues Dr. Mark Vangel, Dr. Fa-Hsuan Lin, and Dr. Matti Håmålåinen.

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58. Learning Visual Spatial Pooling by Strong PCA Dimension Reduction

Haruo Hosoya, Aapo Hyvärinen

https://doi.org/10.1162/neco_a_00843 | pp. 1249-1264

In visual modeling, invariance properties of visual cells are often explained by a pooling mechanism, in which outputs of neurons with similar selectivities to some stimulus parameters are integrated so as to gain some extent of invariance to other parameters. For example, the classical energy model of phase-invariant V1 complex cells pools model simple cells preferring similar orientation but different phases. Prior studies, such as independent subspace analysis, have shown that phase-invariance properties of V1 complex cells can be learned from spatial statistics of natural inputs. However, those previous approaches assumed a squaring nonlinearity on the neural outputs to capture energy correlation; such nonlinearity is arguably unnatural from a neurobiological viewpoint but hard to change due to its tight integration into their formalisms. Moreover, they used somewhat complicated objective functions requiring expensive computations for optimization. In this study, we show that visual spatial pooling can be learned in a much simpler way using strong dimension reduction based on principal component analysis. This approach learns to ignore a large part of detailed spatial structure of the input and thereby estimates a linear pooling matrix. Using this framework, we demonstrate that pooling of model V1 simple cells learned in this way, even with nonlinearities other than squaring, can reproduce standard tuning properties of V1 complex cells. For further understanding, we analyze several variants of the pooling model and argue that a reasonable pooling can generally be obtained from any kind of linear transformation that retains several of the first principal components and suppresses the remaining ones. In particular, we show how the classic Wiener filtering theory leads to one such variant.

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59. An Infinite Restricted Boltzmann Machine

Marc-Alexandre Côté, Hugo Larochelle

https://doi.org/10.1162/neco_a_00848 | pp. 1265-1288

We present a mathematical construction for the restricted Boltzmann machine (RBM) that does not require specifying the number of hidden units. In fact, the hidden layer size is adaptive and can grow during training. This is obtained by first extending the RBM to be sensitive to the ordering of its hidden units. Then, with a carefully chosen definition of the energy function, we show that the limit of infinitely many hidden units is well

defined. As with RBM, approximate maximum likelihood training can be performed, resulting in an algorithm that naturally and adaptively adds trained hidden units during learning. We empirically study the behavior of this infinite RBM, showing that its performance is competitive to that of the RBM, while not requiring the tuning of a hidden layer size.

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60. A Single Hidden Layer Feedforward Network with Only One Neuron in the Hidden Layer Can Approximate Any Univariate Function

Namig J. Guliyev, Vugar E. Ismailov

https://doi.org/10.1162/neco_a_00849 | pp. 1289-1304

The possibility of approximating a continuous function on a compact subset of the real line by a feedforward single hidden layer neural network with a sigmoidal activation function has been studied in many papers. Such networks can approximate an arbitrary continuous function provided that an unlimited number of neurons in a hidden layer is permitted. In this note, we consider constructive approximation on any finite interval of [Formula: see text] by neural networks with only one neuron in the hidden layer. We construct algorithmically a smooth, sigmoidal, almost monotone activation function [Formula: see text] providing approximation to an arbitrary continuous function within any degree of accuracy. This algorithm is implemented in a computer program, which computes the value of [Formula: see text] at any reasonable point of the real axis.

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61. 1D Current Source Density (CSD) Estimation in Inverse Theory: A Unified Framework for Higher-Order Spectral Regularization of Quadrature and Expansion-Type CSD Methods

Pascal Kropf, Amir Shmuel

https://doi.org/10.1162/neco_a_00846 | pp. 1305-1355

Estimation of current source density (CSD) from the low-frequency part of extracellular electric potential recordings is an unstable linear inverse problem. To make the estimation possible in an experimental setting where recordings are contaminated with noise, it is necessary to stabilize the inversion. Here we present a unified framework for zero- and higher-order singular-value-decomposition (SVD)-based spectral regularization of 1D (linear) CSD estimation from local field potentials. The framework is based on two general approaches commonly employed for solving inverse problems: quadrature and basis function expansion. We first show that both inverse CSD (iCSD) and kernel CSD (kCSD) fall into the category of basis function expansion methods. We then use these general categories to introduce two new estimation methods, quadrature CSD (qCSD), based on discretizing the CSD integral equation with a chosen quadrature rule, and representer CSD (rCSD), an even-determined basis function expansion method that uses the problem's data kernels (representers) as basis functions. To determine the best candidate methods to use in the analysis of experimental data, we compared the different methods on simulations under three regularization schemes (Tikhonov, tSVD, and dSVD), three regularization parameter selection methods (NCP, L-curve, and GCV), and seven different a priori spatial smoothness constraints on the CSD distribution. This resulted in a comparison of 531 estimation schemes. We evaluated the estimation schemes according to their source reconstruction accuracy by testing them using different simulated noise levels, lateral source diameters, and CSD depth profiles. We found that ranking schemes according to the

average error over all tested conditions results in a reproducible ranking, where the top schemes are found to perform well in the majority of tested conditions. However, there is no single best estimation scheme that outperforms all others under all tested conditions. The unified framework we propose expands the set of available estimation methods, provides increased flexibility for 1D CSD estimation in noisy experimental conditions, and allows for a meaningful comparison between estimation schemes.

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62. A Novel Nonparametric Approach for Neural Encoding and Decoding Models of Multimodal Receptive Fields

Rahul Agarwal, Zhe Chen, Fabian Kloosterman, Matthew A. Wilson, Sridevi V. Sarma
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Pyramidal neurons recorded from the rat hippocampus and entorhinal cortex, such as place and grid cells, have diverse receptive fields, which are either unimodal or multimodal. Spiking activity from these cells encodes information about the spatial position of a freely foraging rat. At fine timescales, a neuron's spike activity also depends significantly on its own spike history. However, due to limitations of current parametric modeling approaches, it remains a challenge to estimate complex, multimodal neuronal receptive fields while incorporating spike history dependence. Furthermore, efforts to decode the rat's trajectory in one- or two-dimensional space from hippocampal ensemble spiking activity have mainly focused on spike history-independent neuronal encoding models. In this letter, we address these two important issues by extending a recently introduced nonparametric neural encoding framework that allows modeling both complex spatial receptive fields and spike history dependencies. Using this extended nonparametric approach, we develop novel algorithms for decoding a rat's trajectory based on recordings of hippocampal place cells and entorhinal grid cells. Results show that both encoding and decoding models derived from our new method performed significantly better than state-of-the-art encoding and decoding models on 6 minutes of test data. In addition, our model's performance remains invariant to the apparent modality of the neuron's receptive field.

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63. Regularized Multitask Learning for Multidimensional Log-Density Gradient Estimation

Ikko Yamane, Hiroaki Sasaki, Masashi Sugiyama
https://doi.org/10.1162/neco_a_00844 | pp. 1388-1410

Log-density gradient estimation is a fundamental statistical problem and possesses various practical applications such as clustering and measuring nongaussianity. A naive two-step approach of first estimating the density and then taking its log gradient is unreliable because an accurate density estimate does not necessarily lead to an accurate log-density gradient estimate. To cope with this problem, a method to directly estimate the log-density gradient without density estimation has been explored and demonstrated to work much better than the two-step method. The objective of this letter is to improve the performance of this direct method in multidimensional cases. Our idea is to regard the problem of log-density gradient estimation in each dimension as a task and apply regularized multitask learning to the direct log-density gradient estimator. We experimentally demonstrate the usefulness of the proposed multitask method in log-density gradient estimation and mode-seeking clustering.

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64. Nonlinear Memory Capacity of Parallel Time-Delay Reservoir Computers in the Processing of Multidimensional Signals

Lyudmila Grigoryeva, Julie Henriques, Laurent Larger, Juan■Pablo Ortega

https://doi.org/10.1162/neco_a_00845 | pp. 1411-1451

This letter addresses the reservoir design problem in the context of delay-based reservoir computers for multidimensional input signals, parallel architectures, and real-time multitasking. First, an approximating reservoir model is presented in those frameworks that provides an explicit functional link between the reservoir architecture and its performance in the execution of a specific task. Second, the inference properties of the ridge regression estimator in the multivariate context are used to assess the impact of finite sample training on the decrease of the reservoir capacity. Finally, an empirical study is conducted that shows the adequacy of the theoretical results with the empirical performances exhibited by various reservoir architectures in the execution of several nonlinear tasks with multidimensional inputs. Our results confirm the robustness properties of the parallel reservoir architecture with respect to task misspecification and parameter choice already documented in the literature.

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65. Scalable Semisupervised Functional Neurocartography Reveals Canonical Neurons in Behavioral Networks

E. Paxon Frady, Ashish Kapoor, Eric Horvitz, William B. Kristan

https://doi.org/10.1162/neco_a_00852 | pp. 1453-1497

Large-scale data collection efforts to map the brain are underway at multiple spatial and temporal scales, but all face fundamental problems posed by high-dimensional data and intersubject variability. Even seemingly simple problems, such as identifying a neuron/brain region across animals/subjects, become exponentially more difficult in high dimensions, such as recognizing dozens of neurons/brain regions simultaneously. We present a framework and tools for functional neurocartography-the large-scale mapping of neural activity during behavioral states. Using a voltage-sensitive dye (VSD), we imaged the multifunctional responses of hundreds of leech neurons during several behaviors to identify and functionally map homologous neurons. We extracted simple features from each of these behaviors and combined them with anatomical features to create a rich medium-dimensional feature space. This enabled us to use machine learning techniques and visualizations to characterize and account for intersubject variability, piece together a canonical atlas of neural activity, and identify two behavioral networks. We identified 39 neurons (18 pairs, 3 unpaired) as part of a canonical swim network and 17 neurons (8 pairs, 1 unpaired) involved in a partially overlapping preparatory network. All neurons in the preparatory network rapidly depolarized at the onsets of each behavior, suggesting that it is part of a dedicated rapid-response network. This network is likely mediated by the S cell, and we referenced VSD recordings to an activity atlas to identify multiple cells of interest simultaneously in real time for further experiments. We targeted and electrophysiologically verified several neurons in the swim network and further showed that the S cell is presynaptic to multiple neurons in the preparatory network. This study illustrates the basic framework to map neural activity in high dimensions with large-scale recordings and how to extract the rich information necessary to perform analyses in light of intersubject variability.

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66. The Leaky Integrator with Recurrent Inhibition as a Predictor

Henning U. Voss
https://doi.org/10.1162/neco_a_00859 | pp. 1498-1502

The leaky integrator, the basis for many neuronal models, possesses a negative group delay when a time-delayed recurrent inhibition is added to it. By means of this delay, the leaky integrator becomes a predictor for some frequency components of the input signal. The prediction properties are derived analytically, and an application to a local field potential is provided.

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67. Bayesian Inference and Online Learning in Poisson Neuronal Networks

Yanping Huang, Rajesh P. N. Rao
https://doi.org/10.1162/neco_a_00851 | pp. 1503-1526

Motivated by the growing evidence for Bayesian computation in the brain, we show how a two-layer recurrent network of Poisson neurons can perform both approximate Bayesian inference and learning for any hidden Markov model. The lower-layer sensory neurons receive noisy measurements of hidden world states. The higher-layer neurons infer a posterior distribution over world states via Bayesian inference from inputs generated by sensory neurons. We demonstrate how such a neuronal network with synaptic plasticity can implement a form of Bayesian inference similar to Monte Carlo methods such as particle filtering. Each spike in a higher-layer neuron represents a sample of a particular hidden world state. The spiking activity across the neural population approximates the posterior distribution over hidden states. In this model, variability in spiking is regarded not as a nuisance but as an integral feature that provides the variability necessary for sampling during inference. We demonstrate how the network can learn the likelihood model, as well as the transition probabilities underlying the dynamics, using a Hebbian learning rule. We present results illustrating the ability of the network to perform inference and learning for arbitrary hidden Markov models.

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68. Asymmetry of Neuronal Combinatorial Codes Arises from Minimizing Synaptic Weight Change

Christian Leibold, Mauro M. Monsalve■Mercado
https://doi.org/10.1162/neco_a_00854 | pp. 1527-1552

Synaptic change is a costly resource, particularly for brain structures that have a high demand of synaptic plasticity. For example, building memories of object positions requires efficient use of plasticity resources since objects can easily change their location in space and yet we can memorize object locations. But how should a neural circuit ideally be set up to integrate two input streams (object location and identity) in case the overall synaptic changes should be minimized during ongoing learning? This letter provides a theoretical framework on how the two input pathways should ideally be specified. Generally the model predicts that the information-rich pathway should be plastic and encoded sparsely, whereas the pathway conveying less information should be encoded densely and undergo learning only if a neuronal representation of a novel object has to be established. As an example, we consider hippocampal area CA1, which combines place and object information. The model thereby provides a normative account of hippocampal rate remapping, that is, modulations of place field activity by changes of local cues. It may as well be applicable to other brain areas (such as neocortical layer V) that learn combinatorial codes from multiple input streams.

69. Clique-Based Neural Associative Memories with Local Coding and Precoding

Asieh Abolpour Mofrad, Matthew G. Parker, Zahra Ferdosi, Mohammad Hesam Tadayon

https://doi.org/10.1162/neco_a_00856 | pp. 1553-1573

Techniques from coding theory are able to improve the efficiency of neuroinspired and neural associative memories by forcing some construction and constraints on the network. In this letter, the approach is to embed coding techniques into neural associative memory in order to increase their performance in the presence of partial erasures. The motivation comes from recent work by Gripon, Berrou, and coauthors, which revisited Willshaw networks and presented a neural network with interacting neurons that partitioned into clusters. The model introduced stores patterns as small-size cliques that can be retrieved in spite of partial error. We focus on improving the success of retrieval by applying two techniques: doing a local coding in each cluster and then applying a precoding step. We use a slightly different decoding scheme, which is appropriate for partial erasures and converges faster. Although the ideas of local coding and precoding are not new, the way we apply them is different. Simulations show an increase in the pattern retrieval capacity for both techniques. Moreover, we use self-dual additive codes over field [Formula: see text], which have very interesting properties and a simple-graph representation.

70. Hermite Functional Link Neural Network for Solving the Van der Pol–Duffing Oscillator Equation

Susmita Mall, Snehashish Chakraverty

https://doi.org/10.1162/neco_a_00858 | pp. 1574-1598

Hermite polynomial-based functional link artificial neural network (FLANN) is proposed here to solve the Van der Pol-Duffing oscillator equation. A single-layer hermite neural network (HeNN) model is used, where a hidden layer is replaced by expansion block of input pattern using Hermite orthogonal polynomials. A feedforward neural network model with the unsupervised error backpropagation principle is used for modifying the network parameters and minimizing the computed error function. The Van der Pol-Duffing and Duffing oscillator equations may not be solved exactly. Here, approximate solutions of these types of equations have been obtained by applying the HeNN model for the first time. Three mathematical example problems and two real-life application problems of Van der Pol-Duffing oscillator equation, extracting the features of early mechanical failure signal and weak signal detection problems, are solved using the proposed HeNN method. HeNN approximate solutions have been compared with results obtained by the well known Runge-Kutta method. Computed results are depicted in term of graphs. After training the HeNN model, we may use it as a black box to get numerical results at any arbitrary point in the domain. Thus, the proposed HeNN method is efficient. The results reveal that this method is reliable and can be applied to other nonlinear problems too.

71. Optimal Curiosity-Driven Modular Incremental Slow Feature Analysis

Varun Raj Kompella, Matthew Luciw, Marijn Stollenga, Juergen Schmidhuber

https://doi.org/10.1162/neco_a_00855 | pp. 1599-1662

Consider a self-motivated artificial agent who is exploring a complex environment. Part of the complexity is due to the raw high-dimensional sensory input streams, which the agent needs to make sense of. Such inputs can be compactly encoded through a variety of means; one of these is slow feature analysis (SFA). Slow features encode spatiotemporal regularities, which are information-rich explanatory factors (latent variables) underlying the high-dimensional input streams. In our previous work, we have shown how slow features can be learned incrementally, while the agent explores its world, and modularly, such that different sets of features are learned for different parts of the environment (since a single set of regularities does not explain everything). In what order should the agent explore the different parts of the environment? Following Schmidhuber's theory of artificial curiosity, the agent should always concentrate on the area where it can learn the easiest-to-learn set of features that it has not already learned. We formalize this learning problem and theoretically show that, using our model, called curiosity-driven modular incremental slow feature analysis, the agent on average will learn slow feature representations in order of increasing learning difficulty, under certain mild conditions. We provide experimental results to support the theoretical analysis.

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72. A Quasi-Likelihood Approach to Nonnegative Matrix Factorization

Karthik Devarajan, Vincent C. K. Cheung

https://doi.org/10.1162/neco_a_00853 | pp. 1663-1693

A unified approach to nonnegative matrix factorization based on the theory of generalized linear models is proposed. This approach embeds a variety of statistical models, including the exponential family, within a single theoretical framework and provides a unified view of such factorizations from the perspective of quasi-likelihood. Using this framework, a family of algorithms for handling signal-dependent noise is developed and its convergence proved using the expectation-maximization algorithm. In addition, a measure to evaluate the goodness of fit of the resulting factorization is described. The proposed methods allow modeling of nonlinear effects using appropriate link functions and are illustrated using an application in biomedical signal processing.

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73. Credible Intervals for Precision and Recall Based on aK-Fold Cross-Validated Beta Distribution

Yu Wang, Jihong Li

https://doi.org/10.1162/neco_a_00857 | pp. 1694-1722

In typical machine learning applications such as information retrieval, precision and recall are two commonly used measures for assessing an algorithm's performance. Symmetrical confidence intervals based on K-fold cross-validated t distributions are widely used for the inference of precision and recall measures. As we confirmed through simulated experiments, however, these confidence intervals often exhibit lower degrees of confidence, which may easily lead to liberal inference results. Thus, it is crucial to construct faithful confidence (credible) intervals for precision and recall with a high degree of confidence and a short interval length. In this study, we propose two posterior credible intervals for precision and recall based on K-fold cross-validated beta distributions. The first credible interval for precision (or recall) is constructed based on the beta posterior distribution inferred by all K data sets corresponding to K confusion matrices from a K-fold cross-validation. Second, considering that each data set corresponding to a confusion matrix from a K-fold cross-validation can be used to infer a beta posterior distribution of precision (or recall), the second

proposed credible interval for precision (or recall) is constructed based on the average of K beta posterior distributions. Experimental results on simulated and real data sets demonstrate that the first credible interval proposed in this study almost always resulted in degrees of confidence greater than 95%. With an acceptable degree of confidence, both of our two proposed credible intervals have shorter interval lengths than those based on a corrected K-fold cross-validated t distribution. Meanwhile, the average ranks of these two credible intervals are superior to that of the confidence interval based on a K-fold cross-validated t distribution for the degree of confidence and are superior to that of the confidence interval based on a corrected K-fold cross-validated t distribution for the interval length in all 27 cases of simulated and real data experiments. However, the confidence intervals based on the K-fold and corrected K-fold cross-validated t distributions are in the two extremes. Thus, when focusing on the reliability of the inference for precision and recall, the proposed methods are preferable, especially for the first credible interval.

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74. Learning Minimal Latent Directed Information Polytrees

Jalal Etesami, Negar Kiyavash, Todd P. Coleman

https://doi.org/10.1162/neco_a_00874 | pp. 1723-1768

We propose an approach for learning latent directed polytrees as long as there exists an appropriately defined discrepancy measure between the observed nodes. Specifically, we use our approach for learning directed information polytrees where samples are available from only a subset of processes. Directed information trees are a new type of probabilistic graphical models that represent the causal dynamics among a set of random processes in a stochastic system. We prove that the approach is consistent for learning minimal latent directed trees. We analyze the sample complexity of the learning task when the empirical estimator of mutual information is used as the discrepancy measure.

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75. Dynamic Multiscale Modes of Resting State Brain Activity Detected by Entropy Field Decomposition

Lawrence R. Frank, V. L. Galinsky

https://doi.org/10.1162/neco_a_00871 | pp. 1769-1811

The ability of functional magnetic resonance imaging (FMRI) to noninvasively measure fluctuations in brain activity in the absence of an applied stimulus offers the possibility of discerning functional networks in the resting state of the brain. However, the reconstruction of brain networks from these signal fluctuations poses a significant challenge because they are generally nonlinear and nongaussian and can overlap in both their spatial and temporal extent. Moreover, because there is no explicit input stimulus, there is no signal model with which to compare the brain responses. A variety of techniques have been devised to address this problem, but the predominant approaches are based on the presupposition of statistical properties of complex brain signal parameters, which are unprovable but facilitate the analysis. In this article, we address this problem with a new method, entropy field decomposition, for estimating structure within spatiotemporal data. This method is based on a general information field-theoretic formulation of Bayesian probability theory incorporating prior coupling information that allows the enumeration of the most probable parameter configurations without the need for unjustified statistical assumptions. This approach facilitates the construction of brain activation modes directly from the spatial-temporal correlation structure of the data. These modes and their associated spatial-temporal correlation structure can then be used to generate

space-time activity probability trajectories, called functional connectivity pathways, which provide a characterization of functional brain networks.

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76. Active Inference and Learning in the Cerebellum

Karl Friston, Ivan Herreros

https://doi.org/10.1162/neco_a_00863 | pp. 1812-1839

This letter offers a computational account of Pavlovian conditioning in the cerebellum based on active inference and predictive coding. Using eyeblink conditioning as a canonical paradigm, we formulate a minimal generative model that can account for spontaneous blinking, startle responses, and (delay or trace) conditioning. We then establish the face validity of the model using simulated responses to unconditioned and conditioned stimuli to reproduce the sorts of behavior that are observed empirically. The scheme's anatomical validity is then addressed by associating variables in the predictive coding scheme with nuclei and neuronal populations to match the (extrinsic and intrinsic) connectivity of the cerebellar (eyeblink conditioning) system. Finally, we try to establish predictive validity by reproducing selective failures of delay conditioning, trace conditioning, and extinction using (simulated and reversible) focal lesions. Although rather metaphorical, the ensuing scheme can account for a remarkable range of anatomical and neurophysiological aspects of cerebellar circuitry-and the specificity of lesion-deficit mappings that have been established experimentally. From a computational perspective, this work shows how conditioning or learning can be formulated in terms of minimizing variational free energy (or maximizing Bayesian model evidence) using exactly the same principles that underlie predictive coding in perception.

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77. A Mathematical Framework for Statistical Decision Confidence

Balázs Hangya, Joshua I. Sanders, Ádám Kepecs

https://doi.org/10.1162/neco_a_00864 | pp. 1840-1858

Decision confidence is a forecast about the probability that a decision will be correct. From a statistical perspective, decision confidence can be defined as the Bayesian posterior probability that the chosen option is correct based on the evidence contributing to it. Here, we used this formal definition as a starting point to develop a normative statistical framework for decision confidence. Our goal was to make general predictions that do not depend on the structure of the noise or a specific algorithm for estimating confidence. We analytically proved several interrelations between statistical decision confidence and observable decision measures, such as evidence discriminability, choice, and accuracy. These interrelationships specify necessary signatures of decision confidence in terms of externally quantifiable variables that can be empirically tested. Our results lay the foundations for a mathematically rigorous treatment of decision confidence that can lead to a common framework for understanding confidence across different research domains, from human and animal behavior to neural representations.

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78. Linking Neuromodulated Spike-Timing Dependent Plasticity with the Free-Energy Principle

Takuya Isomura, Koji Sakai, Kiyoshi Kotani, Yasuhiko Jimbo

https://doi.org/10.1162/neco_a_00862 | pp. 1859-1888

The free-energy principle is a candidate unified theory for learning and memory in the brain that predicts that neurons, synapses, and neuromodulators work in a manner that minimizes free energy. However, electrophysiological data elucidating the neural and synaptic bases for this theory are lacking. Here, we propose a novel theory bridging the information-theoretical principle with the biological phenomenon of spike-timing dependent plasticity (STDP) regulated by neuromodulators, which we term mSTDP. We propose that by integrating an mSTDP equation, we can obtain a form of Friston's free energy (an information-theoretical function). Then we analytically and numerically show that dopamine (DA) and noradrenaline (NA) influence the accuracy of a principal component analysis (PCA) performed using the mSTDP algorithm. From the perspective of free-energy minimization, these neuromodulatory changes alter the relative weighting or precision of accuracy and prior terms, which induces a switch from pattern completion to separation. These results are consistent with electrophysiological findings and validate the free-energy principle and mSTDP. Moreover, our scheme can potentially be applied in computational psychiatry to build models of the faulty neural networks that underlie the positive symptoms of schizophrenia, which involve abnormal DA levels, as well as models of the NA contribution to memory triage and posttraumatic stress disorder.

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79. The Geometry of Plasticity-Induced Sensitization in Isoinhibitory Rate Motifs

Gautam Kumar, ShiNung Ching

https://doi.org/10.1162/neco_a_00865 | pp. 1889-1926

A well-known phenomenon in sensory perception is desensitization, wherein behavioral responses to persistent stimuli become attenuated over time. In this letter, our focus is on studying mechanisms through which desensitization may be mediated at the network level and, specifically, how sensitivity changes arise as a function of long-term plasticity. Our principal object of study is a generic isoinhibitory motif: a small excitatory-inhibitory network with recurrent inhibition. Such a motif is of interest due to its overrepresentation in laminar sensory network architectures. Here, we introduce a sensitivity analysis derived from control theory in which we characterize the fixed-energy reachable set of the motif. This set describes the regions of the phase-space that are more easily (in terms of stimulus energy) accessed, thus providing a holistic assessment of sensitivity. We specifically focus on how the geometry of this set changes due to repetitive application of a persistent stimulus. We find that for certain motif dynamics, this geometry contracts along the stimulus orientation while expanding in orthogonal directions. In other words, the motif not only desensitizes to the persistent input, but heightens its responsiveness (sensitizes) to those that are orthogonal. We develop a perturbation analysis that links this sensitization to both plasticity-induced changes in synaptic weights and the intrinsic dynamics of the network, highlighting that the effect is not purely due to weight-dependent disinhibition. Instead, this effect depends on the relative neuronal time constants and the consequent stimulus-induced drift that arises in the motif phase-space. For tightly distributed (but random) parameter ranges, sensitization is quite generic and manifests in larger recurrent E-I networks within which the motif is embedded.

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80. The Enhanced Rise and Delayed Fall of Memory in a Model of Synaptic Integration: Extension to Discrete State Synapses

Terry Elliott

https://doi.org/10.1162/neco_a_00867 | pp. 1927-1984

Integrate-and-express models of synaptic plasticity propose that synapses may act as low-pass filters, integrating synaptic plasticity induction signals in order to discern trends before expressing synaptic plasticity. We have previously shown that synaptic filtering strongly controls destabilizing fluctuations in developmental models. When applied to palimpsest memory systems that learn new memories by forgetting old ones, we have also shown that with binary-strength synapses, integrative synapses lead to an initial memory signal rise before its fall back to equilibrium. Such an initial rise is in dramatic contrast to nonintegrative synapses, in which the memory signal falls monotonically. We now extend our earlier analysis of palimpsest memories with synaptic filters to consider the more general case of discrete state, multilevel synapses. We derive exact results for the memory signal dynamics and then consider various simplifying approximations. We show that multilevel synapses enhance the initial rise in the memory signal and then delay its subsequent fall by inducing a plateau-like region in the memory signal. Such dynamics significantly increase memory lifetimes, defined by a signal-to-noise ratio (SNR). We derive expressions for optimal choices of synaptic parameters (filter size, number of strength states, number of synapses) that maximize SNR memory lifetimes. However, we find that with memory lifetimes defined via mean-first-passage times, such optimality conditions do not exist, suggesting that optimality may be an artifact of SNRs.

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81. Dynamic Signal Tracking in a Simple V1 Spiking Model

Guillaume Lajoie, Lai-Sang Young

https://doi.org/10.1162/neco_a_00868 | pp. 1985-2010

This work is part of an effort to understand the neural basis for our visual system's ability, or failure, to accurately track moving visual signals. We consider here a ring model of spiking neurons, intended as a simplified computational model of a single hypercolumn of the primary visual cortex of primates. Signals that consist of edges with time-varying orientations localized in space are considered. Our model is calibrated to produce spontaneous and driven firing rates roughly consistent with experiments, and our two main findings, for which we offer dynamical explanation on the level of neuronal interactions, are the following. First, we have documented consistent transient overshoots in signal perception following signal switches due to emergent interactions of the E- and I-populations. Second, for continuously moving signals, we have found that accuracy is considerably lower at reversals of orientation than when continuing in the same direction (as when the signal is a rotating bar). To measure performance, we use two metrics, called fidelity and reliability, to compare signals reconstructed by the system to the ones presented and assess trial-to-trial variability. We propose that the same population mechanisms responsible for orientation selectivity also impose constraints on dynamic signal tracking that manifest in perception failures consistent with psychophysical observations.

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82. Energy-Efficient Neuromorphic Classifiers

Daniel Martí, Mattia Rigotti, Mingoo Seok, Stefano Fusi

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Neuromorphic engineering combines the architectural and computational principles of systems neuroscience with semiconductor electronics, with the aim of building efficient and compact devices that mimic the synaptic and neural machinery of the brain. The energy consumptions promised by neuromorphic engineering are extremely low, comparable to those of the nervous system. Until now, however, the neuromorphic approach has been restricted to relatively simple circuits and specialized functions, thereby obfuscating a direct comparison of their energy consumption to that used by conventional von Neumann digital machines solving real-world tasks. Here we show that a recent technology developed by IBM can be leveraged to realize neuromorphic circuits that operate as classifiers of complex real-world stimuli. Specifically, we provide a set of general prescriptions to enable the practical implementation of neural architectures that compete with state-of-the-art classifiers. We also show that the energy consumption of these architectures, realized on the IBM chip, is typically two or more orders of magnitude lower than that of conventional digital machines implementing classifiers with comparable performance. Moreover, the spike-based dynamics display a trade-off between integration time and accuracy, which naturally translates into algorithms that can be flexibly deployed for either fast and approximate classifications, or more accurate classifications at the mere expense of longer running times and higher energy costs. This work finally proves that the neuromorphic approach can be efficiently used in real-world applications and has significant advantages over conventional digital devices when energy consumption is considered.

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83. A Note on Divergences

Xiao Liang

https://doi.org/10.1162/neco_a_00878 | pp. 2045-2062

In many areas of neural computation, like learning, optimization, estimation, and inference, suitable divergences play a key role. In this note, we study the conjecture presented by Amari (2009) and find a counterexample to show that the conjecture does not hold generally. Moreover, we investigate two classes of [Formula: see text]-divergence (Zhang, 2004), weighted f-divergence and weighted [Formula: see text]-divergence, and prove that if a divergence is a weighted f-divergence, as well as a Bregman divergence, then it is a weighted [Formula: see text]-divergence. This result reduces in form to the main theorem established by Amari (2009) when [Formula: see text] [Formula: see text].

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84. Simulation Neurotechnologies for Advancing Brain Research: Parallelizing Large Networks in NEURON

William W. Lytton, Alexandra H. Seidenstein, Salvador Durá-Bernal, Robert A. McDougal, Felix Schürmann, Michael L. Hines

https://doi.org/10.1162/neco_a_00876 | pp. 2063-2090

Large multiscale neuronal network simulations are of increasing value as more big data are gathered about brain wiring and organization under the auspices of a current major research initiative, such as Brain Research through Advancing Innovative Neurotechnologies. The development of these models requires new simulation technologies. We describe here the current use of the NEURON simulator with message passing interface (MPI) for simulation in the domain of moderately large networks on commonly available high-performance computers (HPCs). We discuss the basic layout of such simulations, including the methods of simulation setup, the run-time spike-passing paradigm, and postsimulation data storage and

data management approaches. Using the Neuroscience Gateway, a portal for computational neuroscience that provides access to large HPCs, we benchmark simulations of neuronal networks of different sizes (500-100,000 cells), and using different numbers of nodes (1-256). We compare three types of networks, composed of either Izhikevich integrate-and-fire neurons (I&F), single-compartment Hodgkin-Huxley (HH) cells, or a hybrid network with half of each. Results show simulation run time increased approximately linearly with network size and decreased almost linearly with the number of nodes. Networks with I&F neurons were faster than HH networks, although differences were small since all tested cells were point neurons with a single compartment.

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85. Integrator or Coincidence Detector: A Novel Measure Based on the Discrete Reverse Correlation to Determine a Neuron’s Operational Mode

Jacob Kanev, Achilleas Koutsou, Chris Christodoulou, Klaus Obermayer

https://doi.org/10.1162/neco_a_00875 | pp. 2091-2128

In this letter, we propose a definition of the operational mode of a neuron, that is, whether a neuron integrates over its input or detects coincidences. We complete the range of possible operational modes by a new mode we call gap detection, which means that a neuron responds to gaps in its stimulus. We propose a measure consisting of two scalar values, both ranging from -1 to +1: the neural drive, which indicates whether its stimulus excites the neuron, serves as background noise, or inhibits it; the neural mode, which indicates whether the neuron's response is the result of integration over its input, of coincidence detection, or of gap detection; with all three modes possible for all neural drive values. This is a pure spike-based measure and can be applied to measure the influence of either all or subset of a neuron's stimulus. We derive the measure by decomposing the reverse correlation, test it in several artificial and biological settings, and compare it to other measures, finding little or no correlation between them. We relate the results of the measure to neural parameters and investigate the effect of time delay during spike generation. Our results suggest that a neuron can use several different modes simultaneously on different subsets of its stimulus to enable it to respond to its stimulus in a complex manner.

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86. The Space-Clamped Hodgkin-Huxley System with Random Synaptic Input: Inhibition of Spiking by Weak Noise and Analysis with Moment Equations

Henry C. Tuckwell, Susanne Ditlevsen

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We consider a classical space-clamped Hodgkin-Huxley model neuron stimulated by synaptic excitation and inhibition with conductances represented by Ornstein-Uhlenbeck processes. Using numerical solutions of the stochastic model system obtained by an Euler method, it is found that with excitation only, there is a critical value of the steady-state excitatory conductance for repetitive spiking without noise, and for values of the conductance near the critical value, small noise has a powerfully inhibitory effect. For a given level of inhibition, there is also a critical value of the steady-state excitatory conductance for repetitive firing, and it is demonstrated that noise in either the excitatory or inhibitory processes or both can powerfully inhibit spiking. Furthermore, near the critical value, inverse stochastic resonance was observed when noise was present only in the inhibitory input process. The system of deterministic differential equations for the approximate

first- and second-order moments of the model is derived. They are solved using Runge-Kutta methods, and the solutions are compared with the results obtained by simulation for various sets of parameters, including some with conductances obtained by experiment on pyramidal cells of rat prefrontal cortex. The mean and variance obtained from simulation are in good agreement when there is spiking induced by strong stimulation and relatively small noise or when the voltage is fluctuating at subthreshold levels. In the occasional spike mode sometimes exhibited by spinal motoneurons and cortical pyramidal cells, the assumptions underlying the moment equation approach are not satisfied. The simulation results show that noisy synaptic input of either an excitatory or inhibitory character or both may lead to the suppression of firing in neurons operating near a critical point and this has possible implications for cortical networks. Although suppression of firing is corroborated for the system of moment equations, there seem to be substantial differences between the dynamical properties of the original system of stochastic differential equations and the much larger system of moment equations.

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87. Presynaptic Spontaneous Activity Enhances the Accuracy of Latency Coding

Marie Leváková, Massimiliano Tamborrino, Lubomir Kostal, Petr Lánský

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The time to the first spike after stimulus onset typically varies with the stimulation intensity. Experimental evidence suggests that neural systems use such response latency to encode information about the stimulus. We investigate the decoding accuracy of the latency code in relation to the level of noise in the form of presynaptic spontaneous activity. Paradoxically, the optimal performance is achieved at a nonzero level of noise and suprathreshold stimulus intensities. We argue that this phenomenon results from the influence of the spontaneous activity on the stabilization of the membrane potential in the absence of stimulation. The reported decoding accuracy improvement represents a novel manifestation of the noise-aided signal enhancement.

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88. Learning Polychronous Neuronal Groups Using Joint Weight-Delay Spike-Timing-Dependent Plasticity

Haoqi Sun, Olga Sourina, Guang-Bin Huang

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Polychronous neuronal group (PNG), a type of cell assembly, is one of the putative mechanisms for neural information representation. According to the reader-centric definition, some readout neurons can become selective to the information represented by polychronous neuronal groups under ongoing activity. Here, in computational models, we show that the frequently activated polychronous neuronal groups can be learned by readout neurons with joint weight-delay spike-timing-dependent plasticity. The identity of neurons in the group and their expected spike timing at millisecond scale can be recovered from the incoming weights and delays of the readout neurons. The detection performance can be further improved by two layers of readout neurons. In this way, the detection of polychronous neuronal groups becomes an intrinsic part of the network, and the readout neurons become differentiated members in the group to indicate whether subsets of the group have been activated according to their spike timing. The readout spikes representing this information can be used to analyze how PNGs interact with each other or propagate to downstream

networks for higher-level processing.

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89. Dimensionality-Dependent Generalization Bounds for k-Dimensional Coding Schemes

Tongliang Liu, Dacheng Tao, Dong Xu

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The k-dimensional coding schemes refer to a collection of methods that attempt to represent data using a set of representative k-dimensional vectors and include nonnegative matrix factorization, dictionary learning, sparse coding, k-means clustering, and vector quantization as special cases. Previous generalization bounds for the reconstruction error of the k-dimensional coding schemes are mainly dimensionality-independent. A major advantage of these bounds is that they can be used to analyze the generalization error when data are mapped into an infinite- or high-dimensional feature space. However, many applications use finite-dimensional data features. Can we obtain dimensionality-dependent generalization bounds for k-dimensional coding schemes that are tighter than dimensionality-independent bounds when data are in a finite-dimensional feature space? Yes. In this letter, we address this problem and derive a dimensionality-dependent generalization bound for k-dimensional coding schemes by bounding the covering number of the loss function class induced by the reconstruction error. The bound is of order [Formula: see text], where m is the dimension of features, k is the number of the columns in the linear implementation of coding schemes, and n is the size of sample, [Formula: see text] when n is finite and [Formula: see text] when n is infinite. We show that our bound can be tighter than previous results because it avoids inducing the worst-case upper bound on k of the loss function. The proposed generalization bound is also applied to some specific coding schemes to demonstrate that the dimensionality-dependent bound is an indispensable complement to the dimensionality-independent generalization bounds.

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90. The Functional Segregation and Integration Model: Mixture Model Representations of Consistent and Variable Group-Level Connectivity in fMRI

Nathan W. Churchill, Kristoffer H. Madsen, Morten Mørup

https://doi.org/10.1162/neco_a_00877 | pp. 2250-2290

The brain consists of specialized cortical regions that exchange information between each other, reflecting a combination of segregated (local) and integrated (distributed) processes that define brain function. Functional magnetic resonance imaging (fMRI) is widely used to characterize these functional relationships, although it is an ongoing challenge to develop robust, interpretable models for high-dimensional fMRI data. Gaussian mixture models (GMMs) are a powerful tool for parcellating the brain, based on the similarity of voxel time series. However, conventional GMMs have limited parametric flexibility: they only estimate segregated structure and do not model interregional functional connectivity, nor do they account for network variability across voxels or between subjects. To address these issues, this letter develops the functional segregation and integration model (FSIM). This extension of the GMM framework simultaneously estimates spatial clustering and the most consistent group functional connectivity structure. It also explicitly models network variability, based on voxel- and subject-specific network scaling profiles. We compared the FSIM to standard GMM in a predictive cross-validation framework and examined the importance of different model parameters, using both simulated and experimental resting-state data. The reliability of parcellations is not

significantly altered by flexibility of the FSIM, whereas voxel- and subject-specific network scaling profiles significantly improve the ability to predict functional connectivity in independent test data. Moreover, the FSIM provides a set of interpretable parameters to characterize both consistent and variable aspects functional connectivity structure. As an example of its utility, we use subject-specific network profiles to identify brain regions where network expression predicts subject age in the experimental data. Thus, the FSIM is effective at summarizing functional connectivity structure in group-level fMRI, with applications in modeling the relationships between network variability and behavioral/demographic variables.

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91. Neural Quadratic Discriminant Analysis: Nonlinear Decoding with V1-Like Computation

Marino Pagan, Eero P. Simoncelli, Nicole C. Rust

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Linear-nonlinear (LN) models and their extensions have proven successful in describing transformations from stimuli to spiking responses of neurons in early stages of sensory hierarchies. Neural responses at later stages are highly nonlinear and have generally been better characterized in terms of their decoding performance on prespecified tasks. Here we develop a biologically plausible decoding model for classification tasks, that we refer to as neural quadratic discriminant analysis (nQDA). Specifically, we reformulate an optimal quadratic classifier as an LN-LN computation, analogous to "subunit" encoding models that have been used to describe responses in retina and primary visual cortex. We propose a physiological mechanism by which the parameters of the nQDA classifier could be optimized, using a supervised variant of a Hebbian learning rule. As an example of its applicability, we show that nQDA provides a better account than many comparable alternatives for the transformation between neural representations in two high-level brain areas recorded as monkeys performed a visual delayed-match-to-sample task.

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92. Identification of Stable Spike-Timing-Dependent Plasticity from Spiking Activity with Generalized Multilinear Modeling

Brian S. Robinson, Theodore W. Berger, Dong Song

https://doi.org/10.1162/neco_a_00883 | pp. 2320-2351

Characterization of long-term activity-dependent plasticity from behaviorally driven spiking activity is important for understanding the underlying mechanisms of learning and memory. In this letter, we present a computational framework for quantifying spike-timing-dependent plasticity (STDP) during behavior by identifying a functional plasticity rule solely from spiking activity. First, we formulate a flexible point-process spiking neuron model structure with STDP, which includes functions that characterize the stationary and plastic properties of the neuron. The STDP model includes a novel function for prolonged plasticity induction, as well as a more typical function for synaptic weight change based on the relative timing of input-output spike pairs. Consideration for system stability is incorporated with weight-dependent synaptic modification. Next, we formalize an estimation technique using a generalized multilinear model (GMLM) structure with basis function expansion. The weight-dependent synaptic modification adds a nonlinearity to the model, which is addressed with an iterative unconstrained optimization approach. Finally, we demonstrate successful model estimation on simulated spiking data and show that all model functions can

be estimated accurately with this method across a variety of simulation parameters, such as number of inputs, output firing rate, input firing type, and simulation time. Since this approach requires only naturally generated spikes, it can be readily applied to behaving animal studies to characterize the underlying mechanisms of learning and memory.

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93. Surrogate Data Methods Based on a Shuffling of the Trials for Synchrony Detection: The Centering Issue

Mélisande Albert, Yann Bouret, Magalie Fromont, Patricia Reynaud-Bouret

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We investigate several distribution-free dependence detection procedures, all based on a shuffling of the trials, from a statistical point of view. The mathematical justification of such procedures lies in the bootstrap principle and its approximation properties. In particular, we show that such a shuffling has mainly to be done on centered quantities-that is, quantities with zero mean under independence-to construct correct p-values, meaning that the corresponding tests control their false positive (FP) rate. Thanks to this study, we introduce a method, named permutation UE, which consists of a multiple testing procedure based on permutation of experimental trials and delayed coincidence count. Each involved single test of this procedure achieves the prescribed level, so that the corresponding multiple testing procedure controls the false discovery rate (FDR), and this with as few assumptions as possible on the underneath distribution, except independence and identical distribution across trials. The mathematical meaning of this assumption is discussed, and it is in particular argued that it does not mean what is commonly referred in neuroscience to as cross-trials stationarity. Some simulations show, moreover, that permutation UE outperforms the trial-shuffling of Pipa and Grün (2003) and the MTGAUE method of Tuleau-Malot et al. (2014) in terms of single levels and FDR, for a comparable amount of false negatives. Application to real data is also provided.

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94. Variations on the Theme of Synaptic Filtering: A Comparison of Integrate-and-Express Models of Synaptic Plasticity for Memory Lifetimes

Terry Elliott

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Integrate-and-express models of synaptic plasticity propose that synapses integrate plasticity induction signals before expressing synaptic plasticity. By discerning trends in their induction signals, synapses can control destabilizing fluctuations in synaptic strength. In a feedforward perceptron framework with binary-strength synapses for associative memory storage, we have previously shown that such a filter-based model outperforms other, nonintegrative, "cascade"-type models of memory storage in most regions of biologically relevant parameter space. Here, we consider some natural extensions of our earlier filter model, including one specifically tailored to binary-strength synapses and one that demands a fixed, consecutive number of same-type induction signals rather than merely an excess before expressing synaptic plasticity. With these extensions, we show that filter-based models outperform nonintegrative models in all regions of biologically relevant parameter space except for a small sliver in which all models encode memories only weakly. In this sliver, which model is superior depends on the metric used to gauge memory lifetimes (whether a signal-to-noise ratio or a mean first passage time). After comparing and contrasting these various filter models, we discuss the multiple mechanisms and timescales that underlie

both synaptic plasticity and memory phenomena and suggest that multiple, different filtering mechanisms may operate at single synapses.

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95. On the Analytical Solution of Firing Time for SpikeProp

Simon de Montigny, Beno■■t Mâsse

https://doi.org/10.1162/neco_a_00884 | pp. 2461-2473

Error backpropagation in networks of spiking neurons (SpikeProp) shows promise for the supervised learning of temporal patterns. However, its widespread use is hindered by its computational load and occasional convergence failures. In this letter, we show that the neuronal firing time equation at the core of SpikeProp can be solved analytically using the Lambert W function, offering a marked reduction in execution time over the step-based method used in the literature. Applying this analytical method to SpikeProp, we find that training time per epoch can be reduced by 12% to 56% under different experimental conditions. Finally, this work opens the way for further investigations of SpikeProp's convergence behavior.

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96. Continuous Online Sequence Learning with an Unsupervised Neural Network Model

Yuwei Cui, Subutai Ahmad, Jeff Hawkins

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The ability to recognize and predict temporal sequences of sensory inputs is vital for survival in natural environments. Based on many known properties of cortical neurons, hierarchical temporal memory (HTM) sequence memory recently has been proposed as a theoretical framework for sequence learning in the cortex. In this letter, we analyze properties of HTM sequence memory and apply it to sequence learning and prediction problems with streaming data. We show the model is able to continuously learn a large number of variable order temporal sequences using an unsupervised Hebbian-like learning rule. The sparse temporal codes formed by the model can robustly handle branching temporal sequences by maintaining multiple predictions until there is sufficient disambiguating evidence. We compare the HTM sequence memory with other sequence learning algorithms, including statistical methods-autoregressive integrated moving average; feedforward neural networks-time delay neural network and online sequential extreme learning machine; and recurrent neural networks-long short-term memory and echo-state networks on sequence prediction problems with both artificial and real-world data. The HTM model achieves comparable accuracy to other state-of-the-art algorithms. The model also exhibits properties that are critical for sequence learning, including continuous online learning, the ability to handle multiple predictions and branching sequences with high-order statistics, robustness to sensor noise and fault tolerance, and good performance without task-specific hyperparameter tuning. Therefore, the HTM sequence memory not only advances our understanding of how the brain may solve the sequence learning problem but is also applicable to real-world sequence learning problems from continuous data streams.

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97. Chaotic Resonance in Coupled Inferior Olive Neurons with the Llinás Approach Neuron Model

Sou Nobukawa, H. Nishimura
https://doi.org/10.1162/neco_a_00894 | pp. 2505-2532

It is well known that cerebellar motor control is fine-tuned by the learning process adjusted according to rich error signals from inferior olive (IO) neurons. Schweighofer and colleagues proposed that these signals can be produced by chaotic irregular firing in the IO neuron assembly; such chaotic resonance (CR) was replicated in their computer demonstration of a Hodgkin-Huxley (HH)-type compartment model. In this study, we examined the response of CR to a periodic signal in the IO neuron assembly comprising the Llinás approach IO neuron model. This system involves empirically observed dynamics of the IO membrane potential and is simpler than the HH-type compartment model. We then clarified its dependence on electrical coupling strength, input signal strength, and frequency. Furthermore, we compared the physiological validity for IO neurons such as low firing rate and sustaining subthreshold oscillation between CR and conventional stochastic resonance (SR) and examined the consistency with asynchronous firings indicated by the previous model-based studies in the cerebellar learning process. In addition, the signal response of CR and SR was investigated in a large neuron assembly. As the result, we confirmed that CR was consistent with the above IO neuron's characteristics, but it was not as easy for SR.

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98. The Lamellar Structure of the Brain Fiber Pathways

V. L. Galinsky, Lawrence R. Frank
https://doi.org/10.1162/neco_a_00896 | pp. 2533-2556

We present a quantitative statistical analysis of pairwise crossings for all fibers obtained from whole brain tractography that confirms with high confidence that the brain grid theory (Wedeen et al., 2012a) is not supported by the evidence. The overall fiber tracts structure appears to be more consistent with small angle treelike branching of tracts rather than with near-orthogonal gridlike crossing of fiber sheets. The analysis uses our new method for high-resolution whole brain tractography that is capable of resolving fibers crossing of less than 10 degrees and correctly following a continuous angular distribution of fibers even when the individual fiber directions are not resolved. This analysis also allows us to demonstrate that the whole brain fiber pathway system is very well approximated by a lamellar vector field, providing a concise and quantitative mathematical characterization of the structural connectivity of the human brain.

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99. An Online Structural Plasticity Rule for Generating Better Reservoirs

Subhrajit Roy, Arindam Basu
https://doi.org/10.1162/neco_a_00886 | pp. 2557-2584

In this letter, we propose a novel neuro-inspired low-resolution online unsupervised learning rule to train the reservoir or liquid of liquid state machines. The liquid is a sparsely interconnected huge recurrent network of spiking neurons. The proposed learning rule is inspired from structural plasticity and trains the liquid through formating and eliminating synaptic connections. Hence, the learning involves rewiring of the reservoir connections similar to structural plasticity observed in biological neural networks. The network connections can be stored as a connection matrix and updated in memory by using address event representation (AER) protocols, which are generally employed in neuromorphic systems. On investigating the pairwise separation property, we find that trained liquids provide 1.36 0.18 times more interclass separation while retaining similar intraclass separation as compared to random liquids. Moreover, analysis of the linear separation

property reveals that trained liquids are 2.05 0.27 times better than random liquids. Furthermore, we show that our liquids are able to retain the generalization ability and generality of random liquids. A memory analysis shows that trained liquids have 83.67 5.79 ms longer fading memory than random liquids, which have shown 92.8 5.03 ms fading memory for a particular type of spike train inputs. We also throw some light on the dynamics of the evolution of recurrent connections within the liquid. Moreover, compared to separation-driven synaptic modification', a recently proposed algorithm for iteratively refining reservoirs, our learning rule provides 9.30%, 15.21%, and 12.52% more liquid separations and 2.8%, 9.1%, and 7.9% better classification accuracies for 4, 8, and 12 class pattern recognition tasks, respectively.

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100. A Universal Approximation Theorem for Mixture-of-Experts Models

Hien D. Nguyen, Luke R. Lloyd■Jones, Geoffrey J. McLachlan

https://doi.org/10.1162/neco_a_00892 | pp. 2585-2593

The mixture-of-experts (MoE) model is a popular neural network architecture for nonlinear regression and classification. The class of MoE mean functions is known to be uniformly convergent to any unknown target function, assuming that the target function is from a Sobolev space that is sufficiently differentiable and that the domain of estimation is a compact unit hypercube. We provide an alternative result, which shows that the class of MoE mean functions is dense in the class of all continuous functions over arbitrary compact domains of estimation. Our result can be viewed as a universal approximation theorem for MoE models. The theorem we present allows MoE users to be confident in applying such models for estimation when data arise from nonlinear and nondifferentiable generative processes.

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101. Neural Mechanism to Simulate a Scale-Invariant Future

Karthik H. Shankar, Inder Singh, Marc W. Howard

https://doi.org/10.1162/neco_a_00891 | pp. 2594-2627

Predicting the timing and order of future events is an essential feature of cognition in higher life forms. We propose a neural mechanism to nondestructively translate the current state of spatiotemporal memory into the future, so as to construct an ordered set of future predictions almost instantaneously. We hypothesize that within each cycle of hippocampal theta oscillations, the memory state is swept through a range of translations to yield an ordered set of future predictions through modulations in synaptic connections. Theoretically, we operationalize critical neurobiological findings from hippocampal physiology in terms of neural network equations representing spatiotemporal memory. Combined with constraints based on physical principles requiring scale invariance and coherence in translation across memory nodes, the proposition results in Weber-Fechner spacing for the representation of both past (memory) and future (prediction) timelines. We show that the phenomenon of phase precession of neurons in the hippocampus and ventral striatum correspond to the cognitive act of future prediction.

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102. Derivation of Human Chromatic Discrimination Ability from an Information-Theoretical Notion of Distance in Color Space

María da Fonseca, Inés Samengo

https://doi.org/10.1162/neco_a_00903 | pp. 2628-2655

The accuracy with which humans detect chromatic differences varies throughout color space. For example, we are far more precise when discriminating two similar orange stimuli than two similar green stimuli. In order for two colors to be perceived as different, the neurons representing chromatic information must respond differently, and the difference must be larger than the trial-to-trial variability of the response to each separate color. Photoreceptors constitute the first stage in the processing of color information; many more stages are required before humans can consciously report whether two stimuli are perceived as chromatically distinguishable. Therefore, although photoreceptor absorption curves are expected to influence the accuracy of conscious discriminability, there is no reason to believe that they should suffice to explain it. Here we develop information-theoretical tools based on the Fisher metric that demonstrate that photoreceptor absorption properties explain about 87% of the variance of human color discrimination ability, as tested by previous behavioral experiments. In the context of this theory, the bottleneck in chromatic information processing is determined by photoreceptor absorption characteristics. Subsequent encoding stages modify only marginally the chromatic discriminability at the photoreceptor level.

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103. Efficient Neural Codes That Minimize L_p Reconstruction Error

Zhuo Wang, Alan A. Stocker, Daniel D. Lee

https://doi.org/10.1162/neco_a_00900 | pp. 2656-2686

The efficient coding hypothesis assumes that biological sensory systems use neural codes that are optimized to best possibly represent the stimuli that occur in their environment. Most common models use information-theoretic measures, whereas alternative formulations propose incorporating downstream decoding performance. Here we provide a systematic evaluation of different optimality criteria using a parametric formulation of the efficient coding problem based on the [Formula: see text] reconstruction error of the maximum likelihood decoder. This parametric family includes both the information maximization criterion and squared decoding error as special cases. We analytically derived the optimal tuning curve of a single neuron encoding a one-dimensional stimulus with an arbitrary input distribution. We show how the result can be generalized to a class of neural populations by introducing the concept of a meta-tuning curve. The predictions of our framework are tested against previously measured characteristics of some early visual systems found in biology. We find solutions that correspond to low values of [Formula: see text], suggesting that across different animal models, neural representations in the early visual pathways optimize similar criteria about natural stimuli that are relatively close to the information maximization criterion.

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104. Nonparametric e-Mixture Estimation

Ken Takano, Hideitsu Hino, Shotaro Akaho, Noboru Murata

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This study considers the common situation in data analysis when there are few observations of the distribution of interest or the target distribution, while abundant observations are available from auxiliary distributions. In this situation, it is natural to compensate for the lack of data from the target distribution by using data sets from these auxiliary distributions-in other words, approximating the target distribution in a

subspace spanned by a set of auxiliary distributions. Mixture modeling is one of the simplest ways to integrate information from the target and auxiliary distributions in order to express the target distribution as accurately as possible. There are two typical mixtures in the context of information geometry: the [Formula: see text]- and [Formula: see text]-mixtures. The [Formula: see text]-mixture is applied in a variety of research fields because of the presence of the well-known expectation-maximazation algorithm for parameter estimation, whereas the [Formula: see text]-mixture is rarely used because of its difficulty of estimation, particularly for nonparametric models. The [Formula: see text]-mixture, however, is a well-tempered distribution that satisfies the principle of maximum entropy. To model a target distribution with scarce observations accurately, this letter proposes a novel framework for a nonparametric modeling of the [Formula: see text]-mixture and a geometrically inspired estimation algorithm. As numerical examples of the proposed framework, a transfer learning setup is considered. The experimental results show that this framework works well for three types of synthetic data sets, as well as an EEG real-world data set.

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105. Exponential Stability of Almost Periodic Solutions for Memristor-Based Neural Networks with Distributed Leakage Delays

Changjin Xu, Peiluan Li, Yicheng Pang

https://doi.org/10.1162/neco_a_00895 | pp. 2726-2756

In this letter, we deal with a class of memristor-based neural networks with distributed leakage delays. By applying a new Lyapunov function method, we obtain some sufficient conditions that ensure the existence, uniqueness, and global exponential stability of almost periodic solutions of neural networks. We apply the results of this solution to prove the existence and stability of periodic solutions for this delayed neural network with periodic coefficients. We then provide an example to illustrate the effectiveness of the theoretical results. Our results are completely new and complement the previous studies Chen, Zeng, and Jiang (2014) and Jiang, Zeng, and Chen (2015).

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106. Per-Round Knapsack-Constrained Linear Submodular Bandits

Baosheng Yu, Meng Fang, Dacheng Tao

https://doi.org/10.1162/neco_a_00887 | pp. 2757-2789

Linear submodular bandits has been proven to be effective in solving the diversification and feature-based exploration problem in information retrieval systems. Considering there is inevitably a budget constraint in many web-based applications, such as news article recommendations and online advertising, we study the problem of diversification under a budget constraint in a bandit setting. We first introduce a budget constraint to each exploration step of linear submodular bandits as a new problem, which we call per-round knapsack-constrained linear submodular bandits. We then define an [Formula: see text]-approximation unit-cost regret considering that the submodular function maximization is NP-hard. To solve this new problem, we propose two greedy algorithms based on a modified UCB rule. We prove these two algorithms with different regret bounds and computational complexities. Inspired by the lazy evaluation process in submodular function maximization, we also prove that a modified lazy evaluation process can be used to accelerate our algorithms without losing their theoretical guarantee. We conduct a number of experiments, and the experimental results confirm our theoretical analyses.

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107. Complex Neural Network Models for Time-Varying Drazin Inverse

Xuezhong Wang, Yimin Wei, Predrag S. Stanimirovi■

https://doi.org/10.1162/neco_a_00866 | pp. 2790-2824

Two complex Zhang neural network (ZNN) models for computing the Drazin inverse of arbitrary time-varying complex square matrix are presented. The design of these neural networks is based on corresponding matrix-valued error functions arising from the limit representations of the Drazin inverse. Two types of activation functions, appropriate for handling complex matrices, are exploited to develop each of these networks. Theoretical results of convergence analysis are presented to show the desirable properties of the proposed complex-valued ZNN models. Numerical results further demonstrate the effectiveness of the proposed models.

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108. Pattern Completion in Symmetric Threshold-Linear Networks

Carina Curto, Katherine Morrison

https://doi.org/10.1162/neco_a_00869 | pp. 2825-2852

Threshold-linear networks are a common class of firing rate models that describe recurrent interactions among neurons. Unlike their linear counterparts, these networks generically possess multiple stable fixed points (steady states), making them viable candidates for memory encoding and retrieval. In this work, we characterize stable fixed points of general threshold-linear networks with constant external drive and discover constraints on the coexistence of fixed points involving different subsets of active neurons. In the case of symmetric networks, we prove the following antichain property: if a set of neurons [Formula: see text] is the support of a stable fixed point, then no proper subset or superset of [Formula: see text] can support a stable fixed point. Symmetric threshold-linear networks thus appear to be well suited for pattern completion, since the dynamics are guaranteed not to get stuck in a subset or superset of a stored pattern. We also show that for any graph G, we can construct a network whose stable fixed points correspond precisely to the maximal cliques of G. As an application, we design network decoders for place field codes and demonstrate their efficacy for error correction and pattern completion. The proofs of our main results build on the theory of permitted sets in threshold-linear networks, including recently developed connections to classical distance geometry.

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109. Learning Theory Estimates with Observations from General Stationary Stochastic Processes

Hanyuan Hang, Yunlong Feng, Ingo Steinwart, Johan A. K. Suykens

https://doi.org/10.1162/neco_a_00870 | pp. 2853-2889

This letter investigates the supervised learning problem with observations drawn from certain general stationary stochastic processes. Here by general, we mean that many stationary stochastic processes can be included. We show that when the stochastic processes satisfy a generalized Bernstein-type inequality, a unified treatment on analyzing the learning schemes with various mixing processes can be conducted and a sharp oracle inequality for generic regularized empirical risk minimization schemes can be established. The obtained oracle inequality is then applied to derive convergence rates for several learning schemes such as empirical risk minimization (ERM), least squares support vector machines (LS-SVMs) using given generic

kernels, and SVMs using gaussian kernels for both least squares and quantile regression. It turns out that for independent and identically distributed (i.i.d.) processes, our learning rates for ERM recover the optimal rates. For non-i.i.d. processes, including geometrically [Formula: see text]-mixing Markov processes, geometrically [Formula: see text]-mixing processes with restricted decay, [Formula: see text]-mixing processes, and (time-reversed) geometrically [Formula: see text]-mixing processes, our learning rates for SVMs with gaussian kernels match, up to some arbitrarily small extra term in the exponent, the optimal rates. For the remaining cases, our rates are at least close to the optimal rates. As a by-product, the assumed generalized Bernstein-type inequality also provides an interpretation of the so-called effective number of observations for various mixing processes.